

**EQUITY RETURN FORECASTING ON TAIL-RISK-AWARE
PORTFOLIO OPTIMIZATION ON THE NIGERIAN EXCHANGE GROUP
(NGX)**

BY

SANNI ABDUL-AMEEN OLUWADARASIMI

ECN/2021/158

**AN ESSAY SUBMITTED TO THE DEPARTMENT OF ECONOMICS,
FACULTY OF SOCIAL SCIENCES, OBAFEMI AWOLOWO
UNIVERSITY, ILE-IFE, OSUN STATE, NIGERIA.**

**IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
AWARD OF BACHELOR OF SCIENCE (B.Sc. HONS) DEGREE IN
ECONOMICS.**

SEPTEMBER 2026

CERTIFICATION

This is to certify that I, SANNI ABDUL-AMEEN OLUWADARASIMI, with the matriculation number ECN/2021/158 of the Department of Economics, carried out this project for the 2024/2025 academic session, and hence, prepared this project in partial fulfillment of the requirement for the award of Bachelor of Science (B.Sc) Department of Economics, Obafemi Awolowo University, Ile-Ife. Osun State.

Prof O.R Olayeni

Supervisor

Date

Prof O.P Olofin

Head of Department

Date

DEDICATION

This research work is dedicated to Almighty Allah, the source of wisdom, knowledge, and understanding, by whose strength and help I was able to carry out this project. I also dedicate this work to my mum. Mrs Sanni Musilimot Yetunde, whose encouragement and sacrifices have brought me thus far.

ACKNOWLEDGEMENT

I am profoundly grateful to Almighty Allah for His grace, wisdom, and strength which enabled me to successfully complete this research work, and my academic programme in Obafemi Awolowo University, Ile-Ife. My sincere appreciation goes to my supervisor, Prof. O.R Olayeni for his invaluable guidance, insightful feedback, patience and support and even allowing me to embark on this research work. All which contributed immensely to the quality of this work. May Allah continually guide and elevate you, sir. I am also deeply thankful to Prof. D.O Olayungbo for imparting knowledge and offering support throughout my research work. I pray that, the good Lord increases you in knowledge and bless you abundantly.

Special thanks go to my mum, Mrs Sanni Musilimot Yetunde and my sister, Miss Sanni Nofisat Opeyemi for their endless support, encouragement and prayers, which have been my greatest source of motivation. It is my prayer that, your efforts over me will not be in vain. My appreciation also goes out to Mr Samuel for his work which acted as a guidance and standard for how a good undergraduate thesis should be. To my wonderful friends, Iyowun Aishat Iteoluwakiisi, Ojuade Saidat Oluwabukunmi, Olaseyinde John, Possible, Oluwagbemileke, Oluwadara, Taiwo and my co-supervisees, thank you for your support, understanding, and prayers. It is my prayer that, God rewards you bountifully.

ABSTRACT

This study investigates the econometric viability and economic performance of integrating non-linear machine learning return forecasting models with convex Conditional Value-at-Risk (CVaR) portfolio optimization on the Nigerian Exchange Group (NGX). Classical portfolio selection models rooted in Markowitz Mean-Variance Optimization (MVO) operate under the restrictive assumption of Gaussian return distributions and quadratic variance, exposing emerging market investors to severe estimation risk and catastrophic left-tail drawdowns. To address these structural deficiencies, this research deploys a robust quantitative framework evaluating 28 liquid equities from the NGX Pension Index spanning a clean 16-year historical panel from January 2010 through December 2025 (835 weekly observations per asset). Thirteen volume-confirmed technical indicators and autoregressive momentum features were engineered to capture non-linear market interactions without look-ahead bias. Using an Expanding-Window Walk-Forward Validation protocol with 5-fold TimeSeriesSplit cross-validation, Random Forest and eXtreme Gradient Boosting (XGBoost) algorithms were trained and refit quarterly across an out-of-sample test horizon (2021–2025). The resulting dynamic expected return forecasts were incorporated into a coherent linear programming Mean-CVaR optimization model and evaluated against classical MVO, naive 1/N equal weighting, and passive market buy-and-hold indexing across three realistic transaction cost regimes (zero friction, 0.75% institutional PFA brokerage, and 1.50% retail fees). Statistical diagnostic tests (Jarque-Bera and Shapiro-Wilk) conclusively rejected normality across all 28 assets ($p < 0.001$), exhibiting predominantly positive skewness (cross-sectional mean = +0.54) and excess kurtosis across all 28 assets (ranging from 2.20 to 14.47 among assets unaffected by corporate actions), thereby confirming heavy-tailed, leptokurtic distributions and validating the theoretical necessity of CVaR. While raw point prediction errors (RMSE) hover close to historical

baselines due to weekly noise variance, Diebold-Mariano tests confirm that forecast accuracy is statistically indistinguishable from historical baselines across 89.3% of assets ($p > 0.05$). Directional accuracy averaged 38.1% for Random Forest and 37.8% for XGBoost (against 37.9% for the historical baseline across all 28 assets, where below-50% cross-sectional averages reflect zero-return weeks from forward-filling non-trading periods, while reaching 50.8% on liquid large-caps), driven primarily by the Percentage Price Oscillator, Relative Strength Index, and On-Balance Volume. Under zero fee friction, XGBoost + Mean-CVaR achieved a superior gross Sharpe ratio of 1.52 (42.00% annualized return, -22.09% maximum drawdown) compared to 0.91 for the market index and 0.76 for MVO (unadjusted Ledoit-Wolf circular block bootstrap $p = 0.015$, marginal after Benjamini–Hochberg FDR correction at $q = 0.056$ across 18 simultaneous comparisons; +819 basis points Certainty Equivalent Return welfare gain). Crucially, transaction cost sensitivity analysis reveals that the structural CVaR architecture delivers the robustly significant result: under institutional fees, XGB-CVaR's weekly turnover (11.00%) incurs an annual fee drag that renders it significantly inferior to static Historical-CVaR (Sharpe diff -0.25, FDR $q = 0.000$), while unconstrained MVO collapses under turnover drag (-86.38% drawdown, -64.09% CER utility). Low-turnover Historical-CVaR (0.38% weekly turnover) maintains resilient net Sharpe ratios of 1.48 (retail) and 1.49 (institutional). These findings demonstrate that structural tail-risk architecture (CVaR) plays a far more critical role than predictive model complexity in real-world emerging market wealth preservation.

Keywords: Equity Return Forecasting, Machine Learning, Random Forest, XGBoost, Conditional Value-at-Risk (CVaR), Portfolio Optimization, Tail Risk, Nigerian Exchange Group (NGX), Transaction Costs.

TABLE OF CONTENTS

Content	Page
Title Page	i
Certification.....	ii
Dedication	iii
Acknowledgements	iv
Abstract.....	v
Table of Contents	vii
List of Tables	xi
List of Figures.....	xii
CHAPTER ONE: INTRODUCTION.....	1
1.1 Background to the Study	1
1.2 Statement of the Research Problem	7
1.3 Research Questions	10
1.4 Objectives of the Study	11
1.5 Justification of the Study.....	11
1.6 Scope of the Study.....	14
CHAPTER TWO: LITERATURE REVIEW.....	15
2.1 Conceptual Framework	15
2.1.1 Portfolio Optimization.....	15
2.1.2 Tail-Risk-Aware Portfolio Optimization.....	16
2.1.3 Equity Return Forecasting	18
2.1.4 Machine Learning in Financial Forecasting	18
2.1.5 Nigerian Exchange Group (NGX) Market Characteristics.....	19
2.2 Theoretical Framework	21
2.2.1 Modern Portfolio Theory (MPT).....	22
2.2.2 Post Modern Portfolio Theory (PMPT).....	23
2.2.3 Statistical Learning Theory	24
2.3 Empirical Framework.....	27

2.3.1 Mean-Variance Portfolio Optimization	27
2.3.2 VaR and CVaR Based Portfolio Models	28
2.3.3 Integrated Forecasting and Optimization Models	28
2.4 Research Gap.....	31
2.5 Summary of Gaps in Literature	32
CHAPTER THREE: RESEARCH METHODOLOGY	33
3.1 Theoretical Framework	33
3.2 Model Specification	35
3.2.1 Predictive Machine Learning Models.....	35
3.2.2 Random Forest Specification.....	37
3.2.3 eXtreme Gradient Boosting (XGBoost)	39
3.2.4 The Tail-Risk Aware (Mean-CVaR) Optimization Model.....	40
3.3 Estimation Techniques	43
3.3.1 Time-Series Data Partitioning (Train-Test Split)	43
3.3.2 Model Training and Hyperparameter Tuning.....	44
3.3.3 Portfolio Turnover and Transaction Costs.....	44
3.3.4 Feature Importance Extraction	45
3.3.5 Evaluation Metrics for Forecasting and Portfolio Performance	45
3.3.6 Benchmark Portfolios for Evaluation	48
3.4 Description and Measurement of Study Variables.....	49
3.5 Data Preprocessing and Feature Scaling	52
3.6 Software and Implementation Tools	53
3.7 Sources of Data	53
CHAPTER FOUR: DATA ANALYSIS, PRESENTATION AND DISCUSSION	54
4.1 Empirical Results and Asset Universe Breakdown.....	54
4.1.1 Asset Universe Breakdown and Selection Rationale	54
4.2 Descriptive Statistics and Market Characteristics	56
4.3 Statistical Non-Normality and Stationarity Diagnostic Tests	57
4.4 Sectoral Correlation and Multi-Asset Diversification.....	58
4.5 Machine Learning Return Forecasting Results	59
4.5.1 Econometric Evaluation: Forecast Accuracy and Portfolio Value	60
4.6 Feature Importance and Predictive Driver Analysis	63
4.7 Out-of-Sample Portfolio Optimization Backtest and Sensitivity Analysis	64
4.7.1 Statistical Significance of Portfolio Outperformance across Market Fee Regimes	66
4.8 Discussion of Findings	68
CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATIONS	72

5.1 Summary of Findings	72
5.2 Conclusion.....	73
5.3 Policy and Practical Recommendations	74
5.3.1 Regulatory and Macroeconomic Policy Recommendations.....	74
5.3.2 Institutional Asset Allocation Recommendations	74
5.3.3 Quantitative Risk Management Recommendations	75
5.4 Limitations of the Study	75
5.5 Suggestions for Further Research	76
5.6 Contribution to Knowledge.....	77
5.7 Data and Code Availability Statement	77
REFERENCES.....	79

LIST OF TABLES

Table	Page
Table 3.1: Description and Measurement of Study Variables	51
Table 4.0: NGX Pension Index Asset Universe Breakdown & Exclusion Rationale.....	54
Table 4.1: Descriptive Statistics of 28 NGX Equity Log Returns (2010-2025).....	56
Table 4.2: Empirical Non-Normality (JB, SW) and Stationarity (ADF) Diagnostic Tests (All 28 Equities)	57
Table 4.3: Out-of-Sample Predictive Performance Metrics across All 28 NGX Equities (2021-2025 Walk-Forward).....	60
Table 4.4: Out-of-Sample Portfolio Performance with 1.50% Retail Transaction Fees & Market Slippage.....	64
Table 4.5: Out-of-Sample Portfolio Performance under Zero Transaction Fees (Gross Performance - 0.00%)	63
Table 4.5b: Out-of-Sample Portfolio Performance under 0.75% Institutional PFA Transaction Fees	65
Table 4.6: Multi-Tier Pairwise Hypothesis Testing of Sharpe Ratio Equality across Fee Regimes.....	67

LIST OF FIGURES

Figure	Page
Figure 2.1: Conceptual Framework for Equity Return Forecasting and Mean-CVaR Optimization	21
Figure 4.1: NGX 28 Equity Log Return Pairwise Correlation Matrix (2010-2025)	58
Figure 4.2: Feature Importance Breakdown across Technical Indicators.....	63
Figure 4.3: Out-of-Sample Cumulative Wealth Growth Curves (2021-2025)	66

Chapter 1

INTRODUCTION

1.1 Background to the Study

Capital markets play a crucial role in national economic development as they provide the mechanism for listed companies to raise capital and investors to allocate capital to optimal uses (Sulaiman et al., 2023). The stock market - influenced by factors such as demand, supply and investor behavior amongst others, is where listed companies raise capital by selling equity stakes through shares, while investors buy and sell these securities (Nahari, 2025). Investment in the capital markets has been on the rise as one of the many investment vehicles available to individual and institutional investors in recent times. The goal of rational investors in these markets is to minimize risk and maximize return on investments (Mahadevaswamy and Shyamala, 2022), based on their level of risk-aversion when they invest in securities. They share the same goals of diversifying all diversifiable risks and achieving the best possible risk-return ratio (Zaimovic *et al.*, 2024). This connotes the importance of diversification as a major means of sustainable capital appreciation.

Diversification is not an automatic risk-reduction mechanism as its effectiveness depends on the co-movement of the assets returns across markets and over time. When asset returns have low correlation, unsystematic risk of the portfolio can be significantly reduced through diversification. Although diversification is widely used and popular, it still presents a continually evolving challenge in portfolio management - how to optimally allocate resources. For example, international equity diversification was historically more reliable for U.S. investors prior to 2009, however, the post-crisis period favored diversification through a portfolio of stocks and bonds

involving equities, commodities, and government securities. As the correlation among markets increases over time, international portfolio diversification remains if investors continuously measure correlation among these markets so as to create an optimal market combination (Huang *et al.*, 2025).

To deal with the problems that come with diversification, portfolio management has become a step-by-step decision process. The portfolio management includes identifying, selecting and overseeing investments by individuals or institutions with the goal of maximizing returns while minimizing risk (Salo *et al.*, 2024). As markets keep evolving, portfolio management is essential for long-term growth to maximize returns (Yadav *et al.*, 2024). The main goal of portfolio construction is to diversify unsystematic risk through a two-step allocation plan. Firstly, investors balance risk-return by splitting funds between the risk-free assets and the risky assets. After the division, diversification is realized by combining assets with low correlation thereby improving the portfolio's risk-return profile and moving the efficient frontier to better results (Zhang, 2024). Within this framework, the challenge of determining a portfolio composition is an important decision problem which can be solved through portfolio optimization techniques (Salo *et al.*, 2024).

Portfolio optimization is one of the key tenets of investment theory, providing a systematic strategy for dealing with market unpredictability. This offers a systematic framework to help investors find a balance between risk and return and thus maximize their returns on every unit of risk they take. Thus, the portfolio optimization process helps in guiding investors through the uncertainty, volatility, and unpredictability of financial markets, and assists in meeting their unique financial goals (Nahari, 2025). The market downturns that have occurred, including the global financial crisis of 2008 (Global Financial Crisis), were evidence of the inadequacy of traditional diversification methods. Consequently, there has been a greater demand for and usage of models

that allow for a more sophisticated analysis of correlated risk (Senescall & Low, 2024). Optimization models provide a structure for an investor's decision-making and integrate multiple dimensions of risk, including credit, market and liquidity risk, into a single process of making investment decisions rather than relying on traditional methods of diversification (Tjiwidjaja, 2025).

The mean-variance model introduced by Harry Markowitz (1952) is one of the most foundational portfolio optimization models, which related returns and risks of an asset to its expectation and variance of returns which led to the efficient frontier concept (Zsurkis *et al.*, 2024). His foundational contributions laid the foundations for the modern portfolio theory (Mba *et al.*, 2022). By introducing mathematical optimization techniques, the asset allocation problem was formalized into an optimization framework based on the trade-off between expected return and risk. Within this framework, investors may either minimize portfolio risk for a given level of expected return or maximize expected return subject to a specified risk constraint, leading to the derivation of the efficient frontier. The theory is based on various assumptions such as: Homogeneity of investors' expectations, investors' decisions based solely on expected returns and variances, exhibiting rational risk-averse behavior. Financial markets are assumed to be frictionless and efficient with freely divisible assets, unrestricted short selling, and prices that fully reflect available information. Asset returns are uncertain, normally distributed, and correlated, while individual investors are price takers whose trades do not affect market prices (He, 2022).

While Markowitz's mean-variance portfolio theory remains foundational in finance research, it has practical limitations. One key challenge arises from errors in estimating the expected return vector and covariance matrix, which can lead to portfolios that differ substantially from the true optimum (Job, 2022). A second limitation is that MVO does not account for tail risk.

This is because of variance which measures deviation from mean and this can result in huge losses for investors during extreme events (Wang *et al.*, 2024) especially in emerging markets which are marked by high volatility, low liquidity and asymmetric information (Arif *et al.*, 2020). Variance as a risk of measure is also flawed in the sense that it treats both upside and downside deviations equally. This is against the goal of investors as upside deviations are not only welcome but desired and they are mostly concerned with downside risk (Ferrari *et al.*, 2024). This shows that the mean-variance model doesn't truly perceive the investors' perception of risk.

Due to the limitations of the mean and variance approach, other risk measures like Value-at-Risk emerged. Value-at-Risk was particularly introduced since it focuses on assessing the risk based on the highest potential loss that can occur within a specific Confidence Interval (Wang *et al.*, 2024). Value-at-Risk (VaR), a statistical risk measure that aims to quantify potential losses via a specific quantile of the conditional return distribution of a portfolio, has emerged as a common tool for market risk assessment and remains a fundamental piece of risk management practices according to the Basel III Accord (Cheng, 2025; Leccadito *et al.*, 2024).

However, VaR quickly gained acceptance as the standard approach for modeling market risk for several years because of its simplicity. Yet, VaR faces challenges that investors should be aware of. Findings from various studies have shown that during the occurrence of pandemics, VaR models were not accurate at all as far as investors were concerned, even during crises, as noted by Fatouros *et al.* (2023). Moreover, VaR models not only give data concerning potential losses at a certain confidence level, but neglect losses of greater magnitude, as pointed out by (Mozumder *et al.*, 2024). Moreover, it is not sub-additive, meaning the sum of individual asset VaRs can underestimate total portfolio risk. (Leccadito *et al.*, 2024). All of these highlight the need for a risk measure to show severity of extreme losses thereby showing the need for a technique that goes

beyond the traditional measures thereby aiding investors in the construction of a more robust portfolio (Fan, 2025).

Consequently, portfolio optimization models based solely on historical data can misrepresent future risk, indicating the need for methods that incorporate predictive approaches (Ślusarczyk & Ślepaczuk, 2025). Empirical studies highlight the limitations of using historical data for predictive models such as unreliability during economic uncertainty or financial crises, ever-changing market conditions, over fitting and bias of data mining (Zhang, 2025), ineffective capturing of nonlinear and time-varying dynamics (Chung et al., 2025). Historical models have also been observed to do not sufficiently account for volatility features in emerging markets due to poor out-of-sample forecasting (Sahiner, 2022; Adegboyoye & Sarwar, 2025). These limitations highlight the increasing difficulty of accurately forecasting risk and return in unstable and emerging markets, raising important questions about how future portfolio outcomes can be reliably anticipated.

Emerging markets assets returns and risk forecasts face distinct challenging conditions. Volatility in emerging market stocks is high, making it challenging to rely on conventional forecasting models (Alfeus et al., 2024). Conventional forecasting models cannot fit financial time series data because of the stability criteria used in the conventional model, as the financial time series data is prone to stability criteria-related problems because of randomness and randomness-related ingredients of the financial time series data, thereby emphasizing the demand for increased precision forecasting models like Machine Learning Models, which have attracted considerable attention and used elaborately in stock market predictions and forecasting (Ashrafzadeh et al., 2025). Various empirical research works indicate the efficiency and applicability of ML Models in understanding the complex patterns and dynamics present in financial time series data, leading

to increased precision and profits in stock market investments (Fan, 2025; Ashrafzadeh et al., 2025; Kevin & Yugopuspito, 2025). Hence, forecasting models not only assist in estimating predictable risks and rewards but are also helpful in understanding the importance of risk-aware optimization techniques.

Moreover, due to the various benefits in using predictive models, the forecasts can be plugged into optimization techniques - a largely untapped potential to improve outcomes (Zhang, 2024). Machine learning models improve the quality of inputs estimates which helps investors in making better portfolio decisions (Martínez-Barbero et al., 2024). In emerging markets marked by volatility, structural deficiencies and noisy data, ML models have been applied to stock prices forecasting, predicting market crashes, and capturing non-linear relationships in financial time series data (Naeem et al., 2024; Ashrafzadeh et al., 2025). Integrating these forecasts into portfolio optimization frameworks can improve allocation decisions and better minimize portfolio risk (Kevin & Yugopuspito, 2025; Uzoaga et al., 2025a).

Despite several decades of progress in portfolio optimization from the mean variance model in 1952, several challenges are found within and around existing models. The conventional models for mean variance are flawed due to error in estimation with symmetry between upside and downside risks. Furthermore, Value-at-Risk models are inefficient as they are unable to determine the extent or value of tail losses and demonstrate unsteady forecasts using actual historic data for volatile and developing markets (Job, 2022; Wang et al., 2024; Arif et al., 2020; Mozumder et al., 2024). The above factors illustrate and exemplify gaps and needs for improved portfolio optimization tools with accurate and consistent results and recommendations (Ślusarczyk & Ślepaczuk, 2025). The intersection and fusion between predictive models such as machine learning models and Tail-Risk aware models for portfolio optimization, recommend and express a positive

potential for improvement within portfolio and risk management decisions in volatile markets and developing markets (Ashrafzadeh et al., 2025; Kevin and Yugopuspito, 2025), and accordingly underlay and serve as background for this research work and project.

1.2 Statement of the Research Problem

Portfolio optimization is a core mechanism in the markets for balancing risk and return. The popularity arose due to the increased size of the world economy providing investors with an ever-increasing variety of investment opportunities (Li, 2023). This is seen on Nigerian Exchange Group (NGX) - one of Africa's leading financial hubs, where the number of securities listed has increased from 19 - at inception in 1961 to 393 listed securities among which 151 were listed companies as at September 2024 (Nigerian Exchange Group [NGX], 2024). Portfolio optimization research remained a significant concern for researchers for efficient allocation of capital, which entails the allocation of capital based on market fluctuations (Espiga-Fernández et al., 2024). The nature of financial markets, which entails market volatility, vagueness, and market shifts, hampers the predictability of market fluctuations, as seen on the NGX (Uzoaga et al., 2025a). Nevertheless, in efforts to provide a suitable approach, there is a recent emphasis on the integration of predictive models in analytics for a better approach for non-linear analysis, and market shifts (Alfzari et al., 2025). Whether the current approach in portfolio optimization can provide a suitable approach for market risk remains a concern.

In theory, portfolio optimization enables investors to construct portfolios that maximize expected returns for a given level of risk, or equivalently, minimize risk for a given return (Espiga-Fernández *et al.*, 2024). This is done by allocating capital across multiple assets, balancing correlations and individual assets risk to achieve an efficient portfolio (Zsurkis *et al.*, 2024). By combining uncorrelated assets, investors could diversify away all unsystematic risk potentially

improving risk-adjusted returns (Zaimovic *et al.*, 2024). However, to achieve these theoretical benefits depends critically on accurate estimates of returns, covariance, stock forecasts and other input parameters (Zhang, 2025; Ashrafzadeh *et al.*, 2025). However, misestimation of these inputs can distort the portfolio's risk-return profile and investment outcomes (Job, 2022; Wang *et al.*, 2024; Arif *et al.*, 2020). This suggests that these advantages critically depend on how risk and returns are measured and anticipated.

However, in periods of crises, extreme events and recessions such as the 2020 Covid-19 pandemic, 2008 financial crisis, the classical optimization models proved to be inefficient (Fatouros *et al.*, 2023). Empirical studies also show that post-2009 diversification delivers less benefits as correlations among assets rise (Huang *et al.*, 2025). Traditional measures underestimate extreme losses such as Value-at-risk (VaR) which has been criticized not to show the magnitude of extreme losses and not sub-additive (Mozumder *et al.*, 2024; Leccadito *et al.*, 2024). In emerging markets, these issues are amplified due to structural issues, illiquidity, and fat tail distributions (Ojo & Okafor, 2024). Nevertheless, the persistence of these challenges has driven research into alternative risk measures and adaptive optimization methods (Lorimer *et al.*, 2024). Recent research has attempted to address these issues through alternative risk measures and time-series forecasting models such as Expected Shortfall (ES), Autoregressive Integrated Moving Average (ARIMA) and generalized autoregressive conditional heteroskedasticity (GARCH) (Ojo & Okafor, 2024; Mozumder *et al.*, 2024). While these studies and recent research are informative, the issues do not stem from the optimization models themselves, but how expected risks and returns are computed.

Portfolio optimization failures are primarily driven by poor estimation of risk and returns. Studies have criticised the mean-variance optimization model for estimation errors in the expected

return vector and covariance matrix (Job, 2022) giving rise to poor out-of-sample performance (Martínez-Barbero et al., 2024). Although they have achieved significant progress, they are still insufficient as forecasts are made from historical-data performance, have stability assumptions, failure to capture non-linear and time varying dynamics (Ślusarczyk & Ślepaczuk, 2025; Chung et al., 2025). This highlights the need for approaches that can better anticipate future market behavior rather than rely solely on past observations.

In empirical applications, portfolio allocations are generally quite sensitive to calculations for estimated risks and returns (Martínez-Barbero et al., 2024; Ślusarczyk & Ślepaczuk, 2025). The portfolio optimization process essentially involves future expectations with respect to returns and risk because portfolio allocation weights determined by portfolio optimization are derived and based on their future expectations and behaviors and are not limited to their averages as determined before (Martínez-Barbero et al., 2024). As illustrated by relevant research (Martínez-Barbero et al., 2024), portfolios formed using future forecasts by LSTM models outperform those formed using historical return data. Implementing these forecasts within overall portfolio optimization models for improved allocation and reduced portfolio risks can indeed provide benefits (Kevin & Yugopuspito, 2025; Uzoaga et al., 2025a).

Portfolio optimization using classical frameworks has been observed to be highly vulnerable during periods of market stress. This arises because commonly used measures focus on using quantile-based losses and do not sufficiently account for the magnitude of losses in the tail of the return distribution. Conditional Value-at-Risk (CVaR), proposed by Rockafellar and Uryasev (2000), handles this problem effectively as it measures the average loss exceeding a given level of risk over a certain horizon thus suitable for dealing with extreme events (Wahid et al., 2025). With this measure of tail-risk rather than variance, it has been shown to be more robust than

before when dealing with capital allocation with uncertainty. Later research has also confirmed that CVaR performs better than previous models on portfolio returns and Sharpe Ratio (Lorimer et al., 2024). Nonetheless, how effective CVaR is essentially depends on how well input values are estimated and can readily become more effective with forecasting models (Zhang, 2025). This is particularly important when dealing with emerging markets such as the Nigerian Exchange Group (NGX), where financial returns have rather high volatility, illiquidity, and fat-tailed distributions. This indicates an important need for a hybrid combination of tail-risk-aware optimization and forward-looking forecasting of risk and return values to enhance portfolio robustness and returns when dealing with such a challenging environment.

1.3 Research Questions

The study aims to answer the following questions:

1. To what extent do non-linear machine learning models (Random Forest and XGBoost) enhance the predictive accuracy and directional forecasting of weekly equity returns on the Nigerian Exchange Group (NGX) relative to historical baseline models?
2. How effectively does the integration of machine learning return forecasts into a Conditional Value-at-Risk (CVaR) framework mitigate portfolio tail risk and downside drawdowns relative to traditional mean-variance optimization and passive indexing benchmarks?
3. What is the comparative risk-adjusted performance and economic welfare of the integrated ML-CVaR portfolio against traditional Markowitz Mean-Variance Optimization and passive indexing benchmarks across varying transaction cost and turnover friction regimes?

1.4 Objectives of the Study

Following the research questions highlighted above, the broad objective of this study is to examine the effect of equity-return forecasting on tail-risk-aware portfolio optimization on the Nigerian Exchange Group (NGX), while the specific objectives are to:

1. Evaluate the out-of-sample predictive accuracy and directional forecasting performance of non-linear machine learning models (Random Forest and XGBoost) relative to historical baseline models across liquid NGX equities.
2. Formulate and estimate a tail-risk-aware Conditional Value-at-Risk (Mean-CVaR) portfolio optimization model integrating forward-looking machine learning return forecasts, and assess its downside risk mitigation relative to traditional Markowitz Mean-Variance Optimization and passive buy-and-hold indexing.
3. Investigate the comparative risk-adjusted performance, turnover dynamics, and net economic welfare of ML-CVaR portfolios across multi-tier transaction cost regimes reflecting institutional and retail market frictions.

1.5 Justification of the Study

In modern finance, portfolio optimization has evolved to become central to investment decision-making both in theory and practice (Espiga-Fernandez et al., 2024). As financial markets in emerging economies become increasingly volatile, complex, and susceptible to macroeconomic shocks, traditional portfolio frameworks have become strained and inadequate to capture the true state of asset markets (Adegboyoye & Sarwar, 2025). Classical frameworks grounded in Markowitz Modern Portfolio Theory rely on the foundational assumption of normally distributed asset returns and use standard variance as a universal proxy for risk. However, empirical returns on emerging exchanges like the Nigerian Exchange Group (NGX) consistently depart from Gaussian normality,

exhibiting severe skewness, heavy tails, and volatility clustering (Mozumder et al., 2024). During systemic stress (such as currency devaluations, monetary policy tightening, and global market contagion), variance-based models severely underestimate downside tail risk, leaving institutional portfolios vulnerable to catastrophic capital drawdowns (Leccadito et al., 2024). Consequently, poor risk allocation under flawed variance assumptions leads to substantial real economic losses and suboptimal capital allocation.

To resolve these persistent theoretical and operational challenges, this study develops and validates a unified, tail-risk-aware portfolio optimization framework that bridges the divide between non-linear return forecasting and convex downside risk minimization on the Nigerian Exchange Group. Historically, financial econometric research has treated asset return forecasting and portfolio optimization as distinct, isolated tasks (Zhang, 2024). Econometric return models frequently rely on restrictive linear and stationary assumptions that fail to capture complex market interactions and structural breaks (Lyu, 2024), while portfolio optimizers depend on static historical sample means that amplify estimation errors and produce extreme, unstable asset weights (Michaud, 1989; Martínez-Barbero et al., 2024). This study bridges this gap by engineering a two-stage quantitative pipeline. In the predictive stage, advanced non-linear machine learning ensembles, specifically Random Forest and eXtreme Gradient Boosting (XGBoost), are deployed within an Expanding-Window Walk-Forward Validation architecture to extract non-linear momentum patterns and volume-confirmed technical signals directly from weekly NGX equity data without imposing restrictive parametric assumptions (Ashrafzadeh et al., 2025). In the optimization stage, these forward-looking return forecasts are directly synthesized into a convex Conditional Value-at-Risk (Mean-CVaR) linear programming model (Rockafellar & Uryasev,

2000), which replaces standard variance with a coherent tail-loss metric that actively isolates and minimizes extreme tail-risk exposure along the empirical efficient frontier.

The empirical findings and architectural design of this study are of paramount importance to a broad spectrum of institutional and market stakeholders across emerging financial markets. Primarily, this research is targeted at institutional asset managers and Pension Fund Administrators (PFAs) who manage vast retirement savings under strict National Pension Commission (PenCom) regulatory mandates and fiduciary duty constraints. Additionally, the study directly addresses the operational needs of quantitative portfolio managers and risk control officers responsible for capital preservation and risk-budgeting in volatile equity environments. Beyond institutional investors, the research provides valuable empirical insights for regulatory and surveillance institutions, including the Securities and Exchange Commission (SEC) and the Nigerian Exchange Group (NGX), as well as financial econometricians and academic researchers investigating asset pricing anomalies, tail-risk behavior, and algorithmic decision systems in African frontier markets.

This integrated research framework delivers substantive practical, economic, and policy benefits to these stakeholders across three primary dimensions:

First, for institutional fund managers and Pension Fund Administrators, the study provides an empirically robust asset allocation strategy that prioritizes downside capital preservation. By explicitly penalizing the magnitude of losses beyond the Value-at-Risk cutoff rather than penalizing symmetric upside gains, the Mean-CVaR model reduces maximum portfolio drawdowns during periods of severe market downturns, safeguarding retirement assets and institutional wealth against permanent capital impairment. Second, for quantitative risk officers and asset managers, this study resolves the critical divide between theoretical model performance and real-world execution. By stress-testing the integrated machine learning portfolios across multi-

tier transaction fee and market slippage regimes (including 0.75% institutional PFA brokerage and 1.50% retail frictions), the research provides realistic net-of-fee performance benchmarks. This rigorously demonstrates the real-world trade-off between ML-driven rebalancing turnover and transaction fee friction in emerging markets. Third, for market regulators, exchange operators, and financial economists, the study establishes an open-source, reproducible empirical benchmark testing the value of machine learning return forecasts and downside risk optimization on the NGX. This offers concrete empirical evidence to inform regulatory risk-weight guidelines, advance macroprudential stress-testing standards, and stimulate the development of advanced algorithmic investment vehicles in African frontier equity markets.

1.6 Scope of the Study

This study examines the effect of equity-return forecasting on tail-risk aware optimization on the Nigerian Exchange Group (NGX). Weekly price data on selected companies in the NGX Pension Index between the period of 2010 - 2025 were utilized for the study. This study could not extend beyond the selected time frame due to unavailability of price data due to the date of companies' listings. Most available data on price data on publicly listed companies were sourced from ng.investing.com which was limited to the timeframe of the study.

CHAPTER TWO

LITERATURE REVIEW

This research will examine the current literature regarding equity return forecasting and tail-risk aware portfolio optimization in relation to Nigerian Exchange Group (NGX). The chapter will be structured into a conceptual framework, a theoretical framework, an empirical literature review, and a summary of the gaps identified in the existing literature.

2.1 Conceptual Framework

Understanding equity return forecasting and tail-risk-aware portfolio optimization has become increasingly important in modern financial decision-making due to rising market volatility and uncertainty. Forecasting improves the accuracy of return estimates while tail-risk enhances portfolio protection against extreme losses. With the growing applications of machine learning techniques and unique characteristics of emerging markets such as the Nigerian Exchange Group (NGX), these concepts have gained increased importance in investment management.

2.1.1 Portfolio Optimization

Portfolio optimization refers to the systematic process of determining the most efficient allocation of assets in an investment portfolio to achieve a specific financial objective. Its primary objective is to construct a portfolio that either maximizes returns given a specific risk or minimizes risk given a specific return. This involves selecting and weighing assets to strike a balance between risk and return. The foundation of modern portfolio optimization was introduced by Markowitz (1952) through the mean-variance framework, which formalized the relationship between risk and return and laid the groundwork for subsequent portfolio management techniques (Salo et al., 2024).

Building from portfolio optimization, an efficient allocation principle is necessary as it focuses on how resources should be distributed across assets to achieve the best possible balance

between risk and return. Achieving efficiency involves strategies such as diversification, which reduces unsystematic risk by combining assets with low correlations, and careful assessment of risk-return trade-offs, which helps investors decide how much risk they are willing to accept for potential gains. Importantly, the efficiency of a portfolio depends on accurate estimation of returns, variances, and covariances, making continuous monitoring and adjustment essential. By following these principles, investors can construct portfolios that not only optimize performance but are also more resilient to market fluctuations (Zhang, 2024; Okafor & Robertson, 2023).

As a result of modern advances in technology and market complexities - increased market volatility, large datasets and emergence of new asset classes, portfolio optimization has significantly evolved from traditional optimization methods - mean-variance model, black-litterman model and value-at-risk (VAR) model to supplementation of traditional methods or them being replaced with more advanced analytics to better handle these challenges in real time (Hsiao, 2025).

2.1.2 Tail-Risk-Aware Portfolio Optimization

Tail risk generally refers to the risk associated with extreme events that occur in the tails of a probability distribution, representing rare but severe losses. In financial markets, return distributions often exhibit fat tails, implying that extreme losses occur more frequently than a standard normal distribution would predict (Rockafellar & Uryasev, 2002; Mozumder et al., 2024). This phenomenon is often characterized by skewness and excess kurtosis (leptokurtosis), where a high kurtosis value indicates that returns are concentrated around the mean but spread significantly toward the tails, signaling a higher likelihood of extreme volatility and outliers (Alim et al., 2024; Bali et al., 2007).

While understanding how crucial tail-risk is, traditional methods like mean-variance model use variance which treats all deviations symmetrically, therefore ineffective in distinguishing between upside and downside risk, limiting their effectiveness in capturing extreme downside outcomes i.e a critical limitation of variance is that it penalizes positive deviations (gains) and negative deviations (losses) equally (Samaniego Alcántar, 2023). This implies that upside volatility is treated as a risk to be minimized, which can lead to suboptimal outcomes for investors who view returns above the mean as desirable rather than dangerous (Zapata Quimbayo & León, 2025; Lorimer et al., 2024). In contrast, downside risk measures-such as semi-variance, Value-at-Risk (VaR), and Conditional Value-at-Risk (CVaR)-are asymmetric metrics. They differentiate between "good" and "bad" volatility by focusing exclusively on returns that fall below a specific target or threshold (Ferrari et al., 2024; Zsurkis et al., 2024). By isolating the negative portion of the return distribution, downside risk aligns more closely with the investor's intuitive perception of risk as the potential for financial loss rather than mere fluctuation (Mozumder et al., 2024).

Tail risk management is necessary in volatile markets such as the Nigerian Exchange Group (NGX) due to the market's characteristic leptokurtosis and fat tails, which render symmetric measures like variance inadequate for capturing extreme downside potential (Uzoaga et al., 2025b). Furthermore, the NGX exhibits significant leverage effects where negative shocks amplify volatility disproportionately compared to positive news, necessitating the adoption of asymmetric models and tail-risk-aware models such as Conditional Value-at-Risk (CVaR) for robust risk assessment during periods of structural instability (Adegboyo & Sarwar, 2025; Rockafellar & Uryasev, 2002).

2.1.3 Equity Return Forecasting

Equity return forecasting refers to the analytical process of estimating the future financial performance or price direction of stock assets to enhance decision-making in portfolio management and risk mitigation (Yu, 2024; Chao, 2024). This practice operates in contrast to the Efficient Market Hypothesis (EMH) and the Random Walk Hypothesis, which postulate that asset prices reflect all available information and fluctuate randomly, rendering prediction impossible (Manogna & Kulkarni, 2025; San, 2025). However, substantial empirical evidence challenges these hypotheses, suggesting that financial time series exhibit predictable behavioral patterns, non-linear correlations, and volatility clustering that can be modeled to generate abnormal returns (Ashrafzadeh et al., 2025; Yu, 2024).

Equity return forecasting fundamentally transforms portfolio optimization by replacing static historical averages with dynamic, forward-looking estimates, thereby allowing optimization models to adapt to non-stationary market conditions (Manogna & Kulkarni, 2025; Yu, 2024). By integrating advanced algorithms such as Long Short-Term Memory (LSTM) networks and machine learning techniques, investors can enhance asset selection and weight allocation, effectively maximize risk-adjusted returns while mitigate the estimation errors common in traditional Mean-Variance frameworks (Ashrafzadeh et al., 2025; Fan, 2025). This predictive integration facilitates active rebalancing and improves portfolio robustness during volatile periods, yielding performance that consistently surpasses passive benchmarks like the equal-weighted or buy-and-hold strategies (Ślusarczyk & Ślepaczuk, 2025; Martínez-Barbero et al., 2024).

2.1.4 Machine Learning in Financial Forecasting

While traditional econometric models such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) have

historically served as the foundation for financial time series analysis, their reliance on linear assumptions often limits their effectiveness in capturing the chaotic and non-linear nature of stock market volatility (San, 2025; Ajiga et al., 2024). Recent studies suggest that financial markets exhibit complex, dynamic behaviors that rigid statistical models struggle to approximate (Zhang, 2025).

Consequently, there has been a paradigm shift toward Machine Learning (ML) and Deep Learning (DL) methodologies, which offer superior flexibility in modeling high-dimensional data and identifying latent non-linear patterns without strict distributional assumptions (Chao, 2024; Gu et al., 2020). Parallel to deep learning, tree-based methods such as Random Forest (RF) and eXtreme Gradient Boosting (XGBoost) have gained traction for their ability to handle noisy data and prevent overfitting through bagging and boosting mechanisms (Moyoweshumba & Seitshiro, 2025; Chen, 2025). The integration of these techniques into portfolio optimization models enhances asset allocation decisions by providing forward-looking returns estimates and improving the balance between risk and return in portfolio performance.

2.1.5 Nigerian Exchange Group (NGX) market characteristics

The Nigerian Exchange Group (NGX), formerly known as the Nigerian Stock Exchange (NSE), operates as a pivotal financial hub in Africa, structured to facilitate capital allocation and wealth generation through the centralized trading of equities and financial liabilities. The market is characterized by significant volatility clustering and structural breaks such as regime shifts caused by external shocks like oil price fluctuations and political instability (Adegboyo & Sarwar, 2025).

A fundamental characteristic of returns on the NGX is their significant deviation from the normal distribution. A stylized fact consistent with the Efficient Market Hypothesis (EMH)

critiques in emerging markets. Empirical analyses of the All-Share Index (ASI) consistently reject the Gaussian assumption, revealing that return distributions are leptokurtic-possessing higher peaks and heavier tails (fat tails) than a normal distribution (Adegboyoye & Sarwar, 2025; Uzoaga et al., 2025b). This implies that extreme market events and outliers occur more frequently in the Nigerian market than standard linear models would predict (Uzoaga et al., 2025b). Furthermore, NGX returns often exhibit positive skewness, indicating that the distribution of returns is not symmetric and may be influenced by occasional large positive gains, although the market remains highly volatile (Uzoaga et al., 2025b).

All these pose investment challenges to investments in Nigerian equities. Historical precedents, such as the 2014 oil price crash and the COVID-19 pandemic, demonstrate how external crises can trigger severe market downturns, disrupt market capitalization, and erase investor wealth (Fapetu et al., 2021). Furthermore, the market is characterized by frequent structural breaks-abrupt shifts in market regimes caused by policy changes or economic shocks-which complicate the application of traditional linear forecasting models and necessitate adaptive risk management strategies (Adegboyoye & Sarwar, 2025).

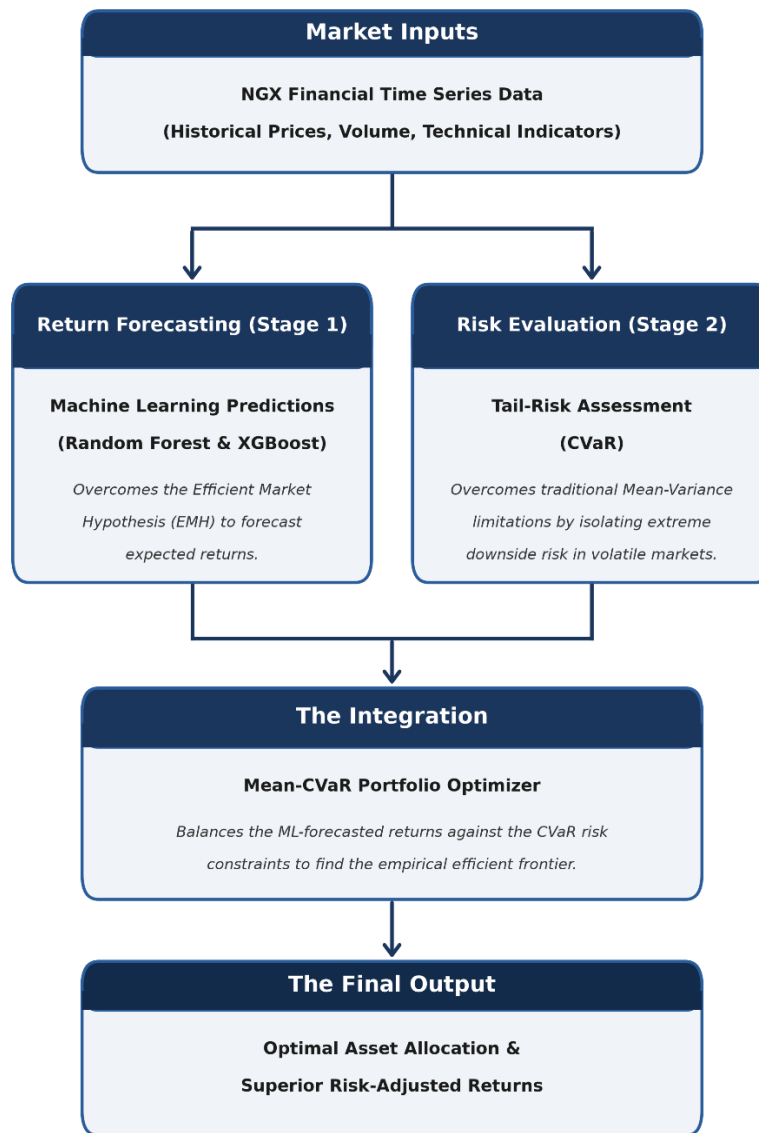


Figure 2.1: Conceptual Framework for Equity Return Forecasting and Mean-CVaR Optimization

2.2 Theoretical Framework

This section examines economic and financial theories that may be beneficial for the study. This subsection provides the theoretical foundation for our research, which can be further utilized in our investigation.

2.2.1 Modern Portfolio Theory (MPT)

Markowitz's (1952) Modern Portfolio Theory (MPT) fundamentally posits that an asset's risk must be evaluated by its contribution to overall portfolio volatility rather than in isolation, demonstrating that diversification reduces unsystematic risk through the combination of assets with imperfect correlations (Job, 2022; Samaniego Alcántar, 2023). This optimization process defines the efficient frontier as a boundary of optimal portfolios that yields the maximum expected return for a defined level of risk or the minimum risk for a target return forming the basis for the mean-variance framework (Chaweewanchon & Chaysiri, 2022; Fan, 2025). Rational investors are thus expected to select portfolios along this frontier that align with their specific risk tolerance, avoiding inefficient allocations that offer inferior risk-return trade-offs (Bodnar et al., 2022; Zsurkis et al., 2024).

Despite its theoretical foundational status, the practical implementation of the Mean-Variance framework is severely compromised by its inherent sensitivity to estimation errors in the input parameters—specifically the expected return vector and the covariance matrix (Martínez-Barbero et al., 2024; Job, 2022). In standard practice, these parameters are unknown and are typically substituted with sample estimates derived from historical data. However, the optimization algorithm treats these estimates as true values, failing to account for parameter uncertainty (Bodnar et al., 2022; Chen, 2025). This leads to a phenomenon where the model acts as an error maximizer, aggressively allocating capital to assets with extreme estimation errors, which results in unstable, corner-solution portfolios that lack diversification (Job, 2022; Salo et al., 2024).

Consequently, portfolios optimized under the traditional Mean-Variance model frequently exhibit poor out-of-sample performance, often failing to outperform simple heuristic benchmarks such as the naive 1/N equal-weighted strategy (Martínez-Barbero et al., 2024; Job, 2022).

2.2.2 Post Modern Portfolio Theory (PMPT)

Post-Modern Portfolio Theory (PMPT) represents a sizable shift in portfolio management by redefining risk not as general volatility, but specifically as the probability of losing money. While early theories treated all price fluctuations equally, PMPT acknowledges that rational investors do not fear unexpected gains; they only fear returns that fall below a specific target or benchmark (Samaniego Alcántar, 2023). This led to the adoption of asymmetric risk measures like semi-variance, which filters out the good volatility and calculates risk based solely on returns that drop below a mean or a required rate of return (Lorimer et al., 2024). By isolating the downside, this approach aligns mathematical modeling more closely with actual investor psychology, where the pain of a loss is often felt more acutely than the pleasure of an equivalent gain (Ferrari et al., 2024).

This focus on asymmetry offers a distinct improvement over the traditional Modern Portfolio Theory (MPT). The classic MPT framework relies on the assumption that asset returns are symmetric in nature, which implies that extreme market crashes are incredibly rare events (Mozumder et al., 2024). However, empirical evidence shows that financial markets frequently exhibit fat tails and skewness, meaning that drastic losses occur far more often than MPT predicts (Adegboyo & Sarwar, 2025). By moving away from standard deviation—which penalizes upside potential just as much as downside loss—PMPT allows for the construction of portfolios that seek to capture growth during bull markets while specifically minimizing exposure to structural breaks and crashes (Chaweewanchon & Chaysiri, 2022).

The theoretical evolution of measuring downside risk acts as a bridge between semi-variance and quantile-based metrics like Value at Risk (VaR) and Conditional Value at Risk (CVaR). While semi-variance looks at the volatility of losses, CVaR (also known as Expected

Shortfall) takes this a step further by estimating the expected magnitude of a loss once a catastrophic threshold has been breached (Alotaibi et al., 2022). Unlike VaR, which simply acts as a cut-off point, CVaR is mathematically coherent, meaning it accounts for the severity of tail events and ensures that diversifying a portfolio does not theoretically increase its risk (Arif et al., 2020). Consequently, modern optimization frameworks have largely transitioned to Mean-CVaR models, which offer a more robust defense against the heavy-tailed risks inherent in volatile markets like the Nigerian Exchange Group (Zapata Quimbayo & León, 2025).

2.2.3 Statistical Learning Theory

Statistical learning theory offers a mathematical basis for understanding how machine learning algorithms estimate functional relationships between variables. Unlike traditional econometrics, which often starts with a rigid, pre-defined structure (like a linear regression equation), machine learning treats the data-generating process as an unknown function that maps input variables—such as past returns or macroeconomic indicators—to an output, like future stock prices (Gu et al., 2020; Chao, 2024). The goal is to use computational power to approximate this complex function by iterating through training data, allowing the model to learn the structure of the data rather than having it imposed by the researcher (Manogna & Kulkarni, 2025). This flexibility allows these models to utilize high-dimensional datasets and extract meaningful signals from noise, a capability that is particularly valuable in financial markets where relationships are non-linear (Zhang, 2025).

A central concept in optimizing these models is the bias-variance tradeoff, which describes the tension between a model's complexity and its predictive stability. If a model is too simple, such as a basic linear regression, it tends to have high bias because it relies on strong assumptions that may not fit the actual market data, essentially missing the underlying pattern (Gu et al., 2020). On

the other hand, if a model is too complex, it exhibits high variance, meaning it is so sensitive that it captures random noise and specific quirks of the training data rather than the true economic signal (Manogna & Kulkarni, 2025). The objective of statistical learning is to find the optimal balance point where the total error-comprising both bias and variance-is minimized, ensuring the model is flexible enough to capture trends but rigid enough to ignore irrelevant noise (Gu et al., 2020).

Closely related to variance is the risk of overfitting, a scenario where a model performs exceptionally well on historical data-essentially memorizing it-but fails to generalize new, unseen data. In the context of financial time series, this occurs when an algorithm mistakes random market fluctuations for predictable patterns, leading to poor performance when applied to real-world trading (Ślusarczyk & Ślepaczuk, 2025; Yu, 2024). To prevent this, researchers employ regularization techniques, which mathematically penalize models for becoming too complex, effectively shrinking the estimated parameters to prevent them from reacting to noise (Gu et al., 2020; Manogna & Kulkarni, 2025). Furthermore, the generalization capability of a model is rigorously tested by splitting data into distinct training, validation, and testing sets, ensuring that the reported accuracy reflects true predictive power rather than in-sample optimization (Ashrafzadeh et al., 2025; Yu, 2024).

The advantage of machine learning lies in its universal approximation property, which allows it to model complex, non-linear data structures that rigid econometric frameworks fail to capture (Gu et al., 2020). While traditional linear models struggle with stylized facts like heavy tails, algorithms such as XGBoost and LSTM autonomously identify latent dependencies in high-dimensional environments (Ashrafzadeh et al., 2025). This flexibility ensures robustness against

shifting market regimes, effectively reducing model misspecification risks compared to static statistical approaches (Manogna & Kulkarni, 2025).

2.3 Empirical Framework

This section of the literature review offers a summary of the connection between portfolio optimization techniques, equity return forecasting and integrated forecasting approach and their individual risk-adjusted return performance.

2.3.1 Mean-Variance Portfolio Optimization

In developed markets, studies show that the MV model doesn't always outperform naive approaches such as an equally weighted portfolio in terms of risk-adjusted returns (like Sharpe ratio). In the US equity market, the naive approach is quite competitive and, in some cases, outperformed where the number of assets is large and unsystematic risk is low (Salo et al., 2024). Additionally, in other studies where unsystematic risk was higher, it is observed to achieve superior returns even during bearish markets showing the mean variance portfolio outshined naive approach due to proper diversification of unsystematic risk (Martínez-Barbero et al., 2024; Ślusarczyk & Ślepaczuk, 2025).

In emerging markets, the MV model didn't perform as well due to issues like high volatility and low liquidity. According to Job (2022), using the MSCI Emerging and Frontier Market Index, the benchmark index outperformed optimization models underscoring the need for better downside risk management and the importance of emerging markets in the world economy.

The failure of the MV model in emerging markets stems from its extreme sensitivity to estimation errors in the input parameters and out-of-sample instability (Bodnar et al., 2022; Rigamonti & Lučivjanská, 2024). Consequently, the reliance on historical sample means and covariances without robustification renders the traditional MV weak and inferior to naive approaches in practical applications (Martínez-Barbero et al., 2024). Based on these studies, it shows a need for more adaptive and downside risk-aware techniques for portfolio optimization.

2.3.2 VaR and CVaR Based Portfolio Models

Value at Risk (VaR) has become a widely accepted benchmark for measuring financial risk, endorsed by the financial regulatory body of Basel Committee on Banking Supervision (BCBS) when it was made a supplement to credit risk measure in the Basel II Accord. VaR is a measure used to determine downside risk when it was initially designed to be used for risk assessment in financial institutions (Arif et al., 2020; Alotaibi et al., 2022; Fatouros et al., 2023).

However, even though it has been used for decades as the standard for measuring tail risk, it lacked efficiency during economic crisis such as the 2020 Covid-19 pandemic and the 2008 global financial crisis. This can be traced to its simplistic assumptions that are summed in the phrase by Einhorn and Brown (2008, as cited in Fatouros et al., 2023), where VaR is compared to "an airbag that works all the time, except when you have a car accident". VaR faces major challenges such as sub-additivity, inadequate percentile of distribution tail loss amongst others (Mozumder et al., 2024; Alotaibi et al., 2022).

Conditional Value at Risk (CVaR) on the other hand acts as a step up since it is derived from VaR but improves coherence properties and satisfies the shortcomings of VaR like subadditivity and convexity properties enabling it to better handle tail risk and achieving a higher accumulated risk-adjusted returns. According to Alotaibi et al. (2022), CVaR achieves superior returns compared to VaR underscoring a more accurate measurement of financial risk.

2.3.3 Integrated Forecasting and Optimization Models

Recent empirical studies have consistently shown that machine learning (ML) models tend to outperform traditional econometric techniques when it comes to forecasting stock returns. Traditional models, such as ARIMA, rely heavily on the assumption that financial data is stable and linear, which often leads to errors because real-world markets are volatile and complex

(Ashrafzadeh et al., 2025). In contrast, advanced algorithms like Random Forest and Neural Networks are designed to identify intricate, non-linear patterns that standard models miss (Gu et al., 2020). For example, San (2025) found that Random Forest models achieved significantly lower error rates (RMSE) compared to the hybrid ARIMA-GARCH model because they could handle market noise and outliers much better. Similarly, Martínez-Barbero et al. (2024) demonstrated that Long Short-Term Memory (LSTM) networks provided much more accurate predictions than traditional statistical methods, both in estimating price values and predicting the direction of the market.

Building superior forecasting, researchers have developed hybrid frameworks that use these machine learning predictions directly into portfolio optimization models. Instead of using historical averages-which assume the future will look like the past-these frameworks use the predicted returns generated by ML models as the inputs for the optimization process (Ashrafzadeh et al., 2025). This method allows the portfolio to adjust dynamically to new information. For instance, Chaweewanchon and Chaysiri (2022) proposed a two-step approach where stocks are first pre-selected based on high predicted returns from a neural network, and then the Markowitz model is used to assign weights. Fan (2025) and Manogna and Kulkarni (2025) utilized similar frameworks, showing that combining LSTM or XGBoost predictions with mean-variance optimization greatly improves how portfolios are constructed by making them more responsive to market changes.

When these forecast-driven frameworks are put to the test, the economic gains are often substantial compared to passive strategies. Gu et al. (2020) provided striking evidence showing that an investor using neural network forecasts to time the S&P 500 could increase their annualized Sharpe ratio from 0.51 (buy-and-hold) to 0.77, essentially earning much higher risk-adjusted

returns. This outperformance holds true even during difficult market periods; Martínez-Barbero et al. (2024) reported that prediction-based portfolios consistently beat the EURO STOXX 50 index, generating positive results even during bear markets. Furthermore, Ślusarczyk and Ślepaczuk (2025) found that strategies utilizing XGBoost forecasts outperformed standard benchmarks in terms of the Information Ratio, confirming that active management based on algorithmic predictions leads to more efficient capital allocation than simple diversification.

In the context of emerging markets like Nigeria, empirical evidence strongly supports the adoption of these hybrid machine learning approaches. Uzoaga et al. (2025a) analyzed the Nigerian Stock Exchange and concluded that Random Forest techniques significantly outperformed linear regression in predicting stock prices, suggesting these tools are well-suited for the Nigerian market structure. Further comparative analysis across Africa by Uzoaga et al. (2025b) revealed that while Random Forest was dominant for Nigerian stock returns, different models like Neural Networks worked better for other markets like Uganda, highlighting the need for model selection. Additionally, a study on the Johannesburg Stock Exchange by Moyoweshumba and Seitshiro (2025) demonstrated that portfolios optimized using XGBoost and Random Forest achieved Sharpe ratios of 1.44 and 1.30 respectively, far surpassing the 0.41 ratio of the traditional Markowitz model. This suggests that hybrid ML strategies hold potential for practical application in some African equity markets, though their out-of-sample efficacy across diverse exchange microstructures warrants empirical scrutiny.

2.4 Research Gap

Despite the growing body of research on portfolio optimization and equity return forecasting, significant gaps remain, particularly in the area of emerging markets like Nigeria. Most of the current studies are focused on developed economies with stable and liquid markets, which often do not capture the volatility, thin trading, and structural shocks characteristic of local markets. This focus limits understanding of how traditional models, such as mean-variance optimization, or even more advanced techniques, behave under conditions of high market uncertainty, leptokurtic returns, and sudden economic or policy shifts.

Furthermore, there is a clear shortage of research integrating predictive modeling with risk-aware portfolio construction. While machine learning and other advanced forecasting techniques have demonstrated strong predictive power, few studies explore how these forecasts can be directly used in optimizing portfolios that account for tail risk. Most research treats forecasting and optimization as separate processes, which reduces the level of application of these methods in real-world decision-making where investors need both accurate predictions and robust allocation strategies simultaneously.

Finally, the practical implementation of these models remains underexplored. Many studies rely on historical data and theoretical assumptions, without testing how forecast-driven, tail-risk-aware portfolios perform under real-world constraints such as transaction costs, liquidity limitations, and market microstructure effects. This leaves a critical gap between theory and practice, creating uncertainty about whether these advanced approaches can truly deliver superior risk-adjusted returns in emerging markets. Addressing these gaps is essential for developing adaptive, resilient, and practically implementable investment strategies tailored to markets like the Nigerian Exchange Group.

2.5 Summary of Gaps in Literature

Overall, the literature shows a lack of research in emerging African markets especially Nigeria that bridges advanced forecasting techniques with robust risk management frameworks. There is limited integration of machine learning predictions into tail-risk-aware portfolio construction, and few studies have tested these methods in real emerging markets. This leaves important questions unanswered about the effectiveness, adaptability, and practical value of such hybrid approaches for investors navigating volatile and complex market environments. These gaps justify the need for a study that develops and evaluates a framework combining predictive forecasting with Conditional Value-at-Risk optimization tailored to Nigerian equities.

CHAPTER THREE

RESEARCH METHODOLOGY

This chapter describes the research methodology employed in the analysis of this study. It entails the theoretical framework, the model specification, the estimation techniques, measurement and description of the variables studied, and lastly, the sources of data.

This chapter is organized as follows. Section 3.1 explains the theoretical framework guiding the conduct of the study, and this is derived from the economic theories specified in Chapter Two. Section 3.2 reveals the model specification used to achieve the objectives of this study, which is in line with the theoretical framework. Section 3.3 comprises the estimation techniques used in the study to achieve objectives. Section 3.4 provides the description and measurement of the study variables. Section 3.4 provides the description and measurement of study variables. Section 3.5 outlines data preprocessing and feature scaling, while Section 3.6 describes the software and computational tools. Section 3.7 concludes the chapter by detailing the sources of data.

3.1 THEORETICAL FRAMEWORK

3.1.1 Modern Portfolio Theory and Tail Risk

As established in Chapter 2, the foundation of portfolio selection lies in Harry Markowitz's Modern Portfolio Theory (MPT), which formalizes the risk-return trade-off by assuming normally distributed returns and relying on symmetric variance as the primary measure of risk (Markowitz, 1952).

However, the empirical reality of emerging markets like the Nigerian Exchange Group (NGX) fundamentally violates these strict assumptions; these markets are instead characterized by

high volatility, low liquidity, asymmetric information, and leptokurtic distributions where extreme downside market events happen much more frequently than normal distributions would predict (Arif et al., 2020). To address these structural violations, Post-Modern Portfolio Theory (PMPT) offers a theoretical solution by redefining risk not as general symmetrical volatility, but specifically as the probability of financial loss, thereby distinguishing between desirable upside gains and detrimental downside drops (Alotaibi et al., 2022). Consequently, this methodology operationalizes PMPT by discarding the traditional mean-variance framework in favor of Conditional Value-at-Risk (CVaR), a coherent risk measure selected to specifically quantify and minimize the average severity of extreme tail losses occurring beyond a catastrophic threshold in the NGX (Rockafellar & Uryasev, 2002).

3.1.2 Machine Learning Theories for Forecasting

To address the limitations of the Efficient Market Hypothesis (EMH) outlined in Chapter 2, this methodology utilizes advanced statistical learning theories. The EMH and the associated Random Walk Hypothesis postulate that asset prices instantaneously reflect all available information and fluctuate randomly, theoretically rendering future price movements unpredictable and active forecasting futile (Fama, 1970).

However, substantial empirical evidence challenges these strict assumptions, particularly in emerging markets like the Nigerian Exchange Group (NGX), which exhibit market frictions, significant volatility clustering, frequent structural breaks, and complex non-linear correlations (Adegboyo & Sarwar, 2025). Statistical learning theories offer a robust mathematical framework that bypasses rigid econometric assumptions, allowing for the extraction of meaningful, latent non-linear dependencies directly from noisy and high-dimensional financial data (Gu et al., 2020).

Consequently, this methodology adopts advanced ensemble machine learning algorithms- specifically Random Forest and eXtreme Gradient Boosting (XGBoost)-to effectively operationalize statistical learning, thereby capturing the complex, non-stationary market patterns that traditional models and the EMH fail to identify (Manogna & Kulkarni, 2025; Ashrafzadeh et al., 2025)

3.2 Model Specification

3.2.1 Predictive Machine Learning Models

The aim of the predictive phase of the study is to forecast future returns of the component assets within the NGX Pension Index. To achieve this, the study frames return forecasting as a supervised multivariate time-series problem. Unlike traditional linear asset pricing models which assume a linear relationship between risk factors and returns, this methodology posits that financial time series are driven by complex, non-linear, and interacting market dynamics.

Let R_{t+1} represent the continuous logarithmic return of a given equity asset for the upcoming week, calculated from the weekly closing prices. The foundational predictive relationship is defined as:

$$R_{t+1} = F(X_t) + \varepsilon_{t+1}$$

where R_{t+1} is the dependent variable representing the weekly logarithmic return at time $t + 1$, F represents the complex, non-linear mapping function that will be empirically approximated by the ensemble machine learning algorithms (Random Forest and XGBoost), X_t is the multi-dimensional feature vector containing the independent variables strictly observed at the end of week t and ε_{t+1} is the idiosyncratic error term (or white noise) representing unobservable market shocks that occur at time $t + 1$, which are independent of the information set at time t .

To provide the machine learning algorithms with a comprehensive representation of market conditions, the feature space is populated with thirteen explicitly selected variables. These variables are derived entirely from the historical Open, High, Low, Close, and Volume (OHLCV) data to capture diverse market dimensions. The feature vector X_t is mathematically defined as a column vector containing the technical momentum, trend, volatility, and volume indicators alongside historical autoregressive return lags.

$$X_t = \begin{bmatrix} MACD_t \\ PPO_t \\ ATR_t \\ RSI_t \\ STOCH_t \\ SMA_t \\ R_t \\ R_{t-1} \\ R_{t-2} \\ R_{t-3} \\ ADX_t \\ SAR_t \\ OBV_t \end{bmatrix}$$

The variables within X_t are categorized into four distinct market dimensions:

1. **Momentum and Oscillator Indicators:** The Moving Average Convergence Divergence ($MACD_t$), Percentage Price Oscillator (PPO_t), Relative Strength Index(RSI_t), and Stochastic Oscillator ($STOCH_t$) measure the velocity and magnitude of recent price swings to identify overbought or oversold conditions.
2. **Autoregressive (Lagged) Returns:** The sequence of $R_{i,t}$, $R_{i,t-1}$, $R_{i,t-2}$, and $R_{i,t-3}$ represents the four most recent weekly logarithmic returns available at prediction time t , feeding historical price momentum directly into the machine learning algorithms.
3. **Trend and Volatility Indicators:** The Simple Moving Average (SMA_t), Average Directional Index (ADX_t), and Stop and Reverse (SAR_t) capture the absolute strength and

direction of the trend. Crucially for the subsequent tail-risk optimization, the Average True Range (ATR_t) captures intra-week market volatility and extreme price fluctuations.

4. **Volume Flow:** On-Balance Volume (OBV_t) acts as a proxy for market liquidity and institutional participation, validating the strength of the observed price trends.

This specific autoregressive functional form is strictly structured to avoid look-ahead bias. By ensuring that the predictive function $F(\cdot)$ maps only the information matrix available at time t to the target return at time $t + 1$, the chronological integrity of the predictive pipeline is preserved, rendering the model suitable for practical portfolio allocation on the Nigerian Exchange Group.

3.2.2 Random Forest Specification

To capture the non-linear interactions among the thirteen technical and momentum indicators, this study employs the Random Forest (RF) algorithm (Breiman, 2001), a robust ensemble machine learning technique.

1. Bootstrap Aggregating (Bagging)

The foundation of the RF model relies on the statistical method of bagging. Given the training dataset of weekly returns and features, the algorithm generates B independent bootstrap samples (random samples drawn with replacement). Let the original training dataset be defined as

$$S = \{(X_1, R_2), (X_2, R_3), \dots, (X_T, R_{T+1})\}.$$

where S is the original historical training dataset of chronological feature-return pairs, X_t is the 13-dimensional technical indicator feature vector observed at weekly time step t , R_{t+1} is the forward realized logarithmic return of the target equity at time step $t + 1$, and T is the total number of historical weekly observations in the training dataset.

For each iteration $b = 1, 2, \dots, B$:

A bootstrap sample S_b is drawn from S . A distinct regression tree, denoted as T_b , is grown using this specific localized sample S_b .

2. Tree Construction and Feature Randomness

At each node of the tree T_b , a random subset of m features is selected from the total $p = 13$ available indicators in the feature vector X_t . For regression tasks, this subset is typically established as $m \approx p/3$.

The algorithm determines the optimal splitting variable j (from the m subset) and split point s by minimizing the Mean Squared Error (MSE) of the resulting partitions. For a given node representing a region of the feature space R_{node} , the split (j, s) is chosen to minimize:

$$\min_{j,s} = \left[\sum_{x_i \in R_{left}(j,s)} (y_i - \hat{y}_{left})^2 + \sum_{x_i \in R_{right}(j,s)} (y_i - \hat{y}_{right})^2 \right]$$

(Where R_{left} and R_{right} are the partitioned feature spaces and $\hat{y}$ represents the mean of the

observed returns within that specific region.

3. The Ensemble Forecast

Each individual tree T_b is grown deeply without pruning. While a single unpruned tree captures highly complex, non-linear market relationships with low bias, it suffers high variance (over-fitting). The Random Forest model neutralizes this variance by averaging the continuous predictive outputs across all B independently trained trees.

The aggregate Random Forest prediction for the weekly return at time $t + 1$, given the specific market information set X_t , is mathematically specified as:

$$R_{t+1}^{RF} = \frac{1}{B} \sum_{b=1}^B T_b(X_t)$$

where R_{t+1}^{RF} is the forecasted logarithmic return of the target asset, B is the total number of regression trees in the forest ensemble (a hyperparameter that will be optimized during the model training phase) and $T_b(X_t)$ is the predicted return of the b -th individual regression tree based on the 13-dimensional feature vector X_t .

3.2.3 eXtreme Gradient Boosting (XGBoost)

The eXtreme Gradient Boosting (XGBoost) algorithm (Chen & Guestrin, 2016) operates as an additive ensemble of classification and regression trees (CART), optimizing a regularized objective function via second-order gradient descent.

1. The Additive Model Specification

The XGBoost algorithm models the expected return at time $t + 1$ as an additive combination of K sequential base learners (regression trees). For a given feature vector X_t containing the thirteen extracted market indicators, the aggregated XGBoost forecast is defined as:

$$\widehat{R_{t+1}^{XGB}} = \sum_{k=1}^K f_k(X_t), \quad f_k \in F$$

where $\widehat{R_{t+1}^{XGB}}$ is the forecasted logarithmic return, K is the total number of additive trees and f_k represents an independent classification and regression tree (CART) structure within the functional space F of all possible trees.

2. The Regularized Objective Function

Financial time series are notoriously noisy, making complex models highly susceptible to overfitting. XGBoost addresses this by optimizing a regularized objective function at each sequential step t (not to be confused with time index t). The algorithm seeks to minimize the following loss function:

$$L = \sum_{i=1}^N l(R_{i,t+1}, \widehat{R_{i,t+1}}) + \sum_{k=1}^K \Omega(f_k)$$

where $l(\cdot)$ is a differentiable convex loss function, specifically the Mean Squared Error (MSE), which measures the difference between the actual observed return $R_{i,t+1}$ and the model's prediction $\widehat{R_{i,t+1}}$ and $\Omega(f_k)$ is the explicit regularization penalty applied to the model's complexity to prevent overfitting the technical indicators.

The regularization term is mathematically defined as:

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda |w|^2$$

where T is the number of terminal nodes (leaves) in the tree, w represents the leaf weights, γ penalizes the addition of new splits (controlling tree depth), and λ is the L2 regularization term on the weights. By aggressively penalizing complexity, the XGBoost model isolates true predictive signals from transient market noise on the NGX.

3.2.4 The Tail-Risk Aware (Mean-CVaR) Optimization Model

The continuous return forecasts generated by the Random Forest and XGBoost algorithms form separate expected return vectors, denoted as μ_{RF} and μ_{XGB} , which serve as primary inputs for the respective portfolio optimization algorithms.

To account for the severe downside volatility inherent in emerging equity markets, this study discards the traditional Mean-Variance framework (which assumes normally distributed returns) in favor of the Mean-Conditional Value at Risk (Mean-CVaR) framework.

1. Formulation of Portfolio Loss and VaR

Let w represent the vector of portfolio weights allocated to the N constituent assets of the NGX Pension Index and r represent the vector of actual asset returns. The portfolio loss function is defined as the negative of the portfolio return:

$$L(w, r) = -w^T r$$

where $L(w, r)$ is the scalar portfolio loss representing the negative portfolio return, $w = [w_1, \dots, w_n]^T$ is the n -dimensional column vector of portfolio asset weights, $r = [r_1, \dots, r_n]^T$ is the n -dimensional vector of asset logarithmic returns, w_i and r_i denote the allocation weight and return of the i -th individual equity, and n is the total number of constituent equities ($n = 28$). Before defining CVaR, the Value at Risk (VaR) at a specified confidence level $\alpha \in (0, 1)$ (e.g., 95%) must be established. VaR represents the minimum loss expected in the $(1 - \alpha)$ worst-case scenarios:

$$VaR_\alpha(w) = \min \{ \gamma \in R : P(L(w, r) \leq \gamma) \geq \alpha \}$$

where $VaR_\alpha(w)$ denotes the Value-at-Risk at confidence level α representing the minimum loss threshold, ζ is the loss threshold parameter, α is the prescribed confidence level ($\alpha = 0.95$ for a 95% confidence horizon), and $\Psi(w, \zeta) = P(L(w, r) \leq \zeta)$ is the cumulative distribution function of portfolio loss.

2. Conditional Value at Risk (CVaR)

Conditional Value-at-Risk (CVaR), also termed Expected Shortfall, measures the expected portfolio loss conditional on losses exceeding the Value-at-Risk threshold. Unlike VaR, which is tail-blind beyond the cutoff percentile, CVaR captures the magnitude of extreme tail losses and satisfies the properties of a coherent risk measure (Artzner et al., 1999). For a continuous loss distribution, CVaR is defined as:

$$CVaR_\alpha(w) = E[L(w, r) \mid L(w, r) \geq VaR_\alpha(w)]$$

where $CVaR_\alpha(w)$ is the Conditional Value-at-Risk (or Expected Shortfall) quantifying the expected loss given that portfolio loss exceeds the $VaR_\alpha(w)$ threshold, α is the confidence level

($\alpha = 0.95$), $p(r)$ is the joint probability density function of equity returns r , and $L(w, r)$ is the portfolio loss function.

Following the formulation by Rockafellar and Uryasev (2000), CVaR can be seamlessly integrated into a linear programming problem by minimizing an auxiliary continuous and convex function:

$$F_\alpha(w, \gamma) = \gamma + \frac{1}{1-\alpha} E[\max(0, L(w, r) - \gamma)]$$

where $F_\alpha(w, \gamma)$ is the continuous, convex auxiliary performance function whose global minimum over γ yields $CVaR_\alpha(w)$, γ is the threshold decision variable representing Value-at-Risk, α is the confidence level ($\alpha = 0.95$), and $[L(w, r) - \gamma]^+ = \max(0, L(w, r) - \gamma)$ is the non-negative loss excess operator.

3. The Optimization Objective

The ultimate objective of the model is to minimize the tail risk (CVaR) while achieving a minimum acceptable expected return driven by the machine learning forecasts. The optimization problem is specified as:

$$\min_{w, \gamma} \left(\gamma + \frac{1}{q(1-\alpha)} \sum_{j=1}^q [-w^T r_j - \gamma]^+ \right)$$

where q is the total number of discrete historical weekly observations (scenarios) in the lookback optimization window, u_k is the non-negative slack variable capturing portfolio loss excess for the k -th historical scenario ($u_k \geq -w^T r_k - \gamma$), γ is the estimated $VaR_\alpha(w)$ threshold, and α is the confidence parameter ($\alpha = 0.95$).

Subject to the following empirical constraints:

1. **Target Return Constraint:** $w^T \hat{\mu} \geq \mu_{target}$ (The portfolio's expected return, leveraging the ML vector $\hat{\mu}$, must meet or exceed a specified target).

2. **Budget Constraint:** $\sum_{i=1}^N w_i = 1$ (The portfolio is fully invested, utilizing 100% of available capital).
3. **Long-Only Constraint:** $w_i \geq 0 \quad \forall i$ (Short selling is prohibited, reflecting institutional realities on the NGX).

By directly feeding the non-linear machine learning predictions ($\hat{\mu}$) into this coherent risk-optimization framework, the final portfolio captures both predictive momentum and robust defense against extreme market downturns.

3.3 Estimation Techniques

The estimation techniques applied in this study are carefully chosen to meet the stated objectives and the nature of data employed. Since the study employs time-series over the period of 2010 -2025, it is important that the employed techniques can pick up the statistical features of the data and the specific research questions

3.3.1 Time-Series Data Partitioning (Train-Test Split)

Evaluating machine learning models requires testing them on unseen, out-of-sample data. While traditional machine learning relies on randomized K-fold cross-validation, applying random splits to financial time-series data introduces *look-ahead bias*. This severe methodological flaw occurs when future market information inadvertently leaks into the training dataset, destroying the chronological order of events, and artificially inflating the model's predictive accuracy.

To preserve temporal order and eliminate look-ahead bias, this study employs an Expanding-Window Walk-Forward Validation protocol. Hyperparameters for Random Forest and XGBoost were tuned once using a 5-fold TimeSeriesSplit cross-validation on the initial 11-year in-sample period (2010–2020, 575 weekly observations) via GridSearchCV. During the out-of-sample evaluation period (2021–2025, 260 weekly observations), models were trained on

expanding historical windows and refit quarterly (every 13 weeks), generating dynamic, one-step-ahead out-of-sample weekly return forecasts.

At each quarterly refitting step t , the training window encompasses all historical weekly data from inception (January 2010) up to t , thereby allowing the machine learning algorithms to continually assimilate evolving market regimes and structural volatility shifts.

3.3.2 Model Training and Hyperparameter Tuning

Hyperparameters were tuned once on the 2010–2020 in-sample period using GridSearchCV with 5-fold TimeSeriesSplit, focusing on parameters that penalize model complexity:

1. **Random Forest Parameters:** The search optimizes the number of trees in the forest ($n_estimators$), the maximum depth of each tree (max_depth), and the minimum number of samples required to split an internal node ($min_samples_split$).
2. **XGBoost Parameters:** The search optimizes the learning rate or step size shrinkage ($learning_rate$), the maximum tree depth (max_depth), and the minimum loss reduction required to make a further partition ($gamma$), which directly controls the structural regularization.

3.3.3 Portfolio Turnover and Transaction Costs

To ensure that the optimized Mean–CVaR portfolio strategy is economically viable and not merely theoretically profitable, it is imperative to account for real–world trading frictions, such as brokerage fees and slippage on the Nigerian Exchange Group (NGX) (Ashrafzadeh et al., 2025). This requires measuring portfolio turnover, which quantifies the absolute change in asset weights from one weekly rebalancing period to the next (Zsurkis et al., 2024). Total rebalancing transaction

costs at week t are mathematically defined as $\text{Cost}_t = c \times \sum |w_{i,t} - w_{i,t-1}|$, where c denotes the proportional transaction fee rate. This study evaluates portfolio performance across three execution friction regimes: a frictionless gross baseline ($c = 0.00\%$), an institutional PFA brokerage scenario ($c = 0.75\%$), and a retail investor friction scenario ($c = 1.50\%$). Net portfolio returns are constructed by deducting these calculated transaction costs from gross returns at each weekly rebalancing step.

3.3.4 Feature Importance Extraction

Due to the criticism of ensemble models of being black boxes in traditional econometrics and to ensure economic interpretability, this study extracts the built-in feature importance scores from the optimized models to determine which of the thirteen technical indicators exert the strongest predictive influence on NGX equity returns.

1. **Random Forest Interpretation:** Importance is calculated using the Mean Decrease in Impurity (Gini Importance). It measures the total reduction of the Mean Squared Error (MSE) brought by that specific feature across all trees in the forest.
2. **XGBoost Interpretation:** Importance is measured using Information Gain. It evaluates the relative contribution of each feature to the model by calculating the improvement in accuracy it provides to the branches it is on.

3.3.5 Evaluation Metrics for Forecasting and Portfolio Performance

The methodology utilizes a dual-evaluation framework to assess both the predictive accuracy of the machine learning algorithms (Stage 1) and the financial viability of the resulting Mean-CVaR optimized portfolio against the Mean-Variance baseline (Stage 2).

1. Predictive Forecasting Metrics

To evaluate the machine learning phase, the deviation of the forecasted expected returns ($\hat{R}$) from the actual observed returns (R) is measured using the continuous quadratic error metric:

1. Root Mean Squared Error (RMSE): Penalizes larger forecasting errors more heavily, which is crucial for identifying severe prediction failures before they are fed into the optimizer.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\mu}_i - R_i)^2}$$

where $RMSE$ is the root mean squared error measuring the quadratic penalty of out-of-sample prediction errors, R_i is the actual realized weekly logarithmic return of the asset at week i , $\hat{\mu}_i$ is the model's forecasted return at week i , and n is the total number of out-of-sample weekly forecasting evaluation steps. To determine whether predictive error differentials between the machine learning models and the historical baseline are statistically significant under quadratic loss, the Diebold–Mariano (1995) test is applied with Newey–West HAC standard errors.

Directional Accuracy (DA): Quantifies the percentage of out-of-sample forecast periods where the predicted return direction matches the realized return direction, evaluating the market timing capability of each forecasting model. Formally, Directional Accuracy is expressed as:

$$DA = \frac{1}{T} \sum_{t=1}^T 1_{\{sign(R_{i,t+1}) = sign(\hat{R}_{i,t+1})\}} \times 100\%$$

2. Risk-Adjusted Portfolio Metrics

To evaluate the final portfolio allocation, performance is measured using risk-adjusted return metrics. This requires a localized risk-free rate proxy (R_f), for which the de-annualized Central Bank of Nigeria (CBN) 91-day Treasury Bill rate is utilized.

1. Sharpe Ratio (Sharpe, 1966): Measures the excess return per unit of total portfolio risk (standard deviation,).

$$Sharpe = \frac{E[R_p] - R_f}{\sigma_p}$$

where *Sharpe* denotes the annualized Sharpe ratio measuring excess return generated per unit of total risk, $E[R_p]$ is the annualized expected return of the portfolio, R_f is the annualized risk-free rate (CBN 91-day Treasury Bill rate), and σ_p is the annualized standard deviation of weekly portfolio returns.

2. Sortino Ratio (Sortino & Price, 1994): Aligning with the thesis's focus on tail risk, this ratio isolates downside volatility. It replaces total standard deviation with downside deviation (), penalizing only the returns that fall below .

$$Sortino = \frac{E[R_p] - R_f}{\sigma_d}$$

where *Sortino* is the downside risk-adjusted return metric, $E[R_p]$ is the annualized expected portfolio return, R_f is the annualized risk-free benchmark rate, and σ_d is the downside semi-deviation isolating return volatility below the risk-free rate.

3. Maximum DrawDown (MDD): Quantifies the portfolio's absolute worst-case scenario. It measures the maximum observed loss from a historical peak to a subsequent trough before a new peak is attained.

$$MDD = \frac{Trough\ Value - Peak\ Value}{Peak\ Value}$$

where *MDD* represents the maximum peak-to-trough percentage loss experienced by the portfolio over the investment horizon, V_t is the cumulative portfolio value at week t , and $\max_{s \leq t} V_s$ is the running historical peak value of the portfolio up to week t .

4. Certainty Equivalent Return (CER): Measures the risk-free rate of return that an investor with risk-aversion coefficient γ would accept in lieu of holding the risky portfolio.

Formally, CER is formulated as:

$$CER = \hat{\mu}_p - \frac{\gamma}{2} \hat{\sigma}_p^2$$

Here $\gamma = 3.0$ is the risk-aversion coefficient, consistent with the MVO utility specification.

By utilizing these metrics, the study empirically validates whether the non-linear machine learning forecasts, combined with CVaR tail-risk constraints, genuinely deliver superior risk-adjusted performance on the Nigerian Exchange. To determine whether empirical differences in portfolio performance are statistically significant, formal econometric hypothesis tests are conducted, including the asymptotic Jobson–Korkie test (Jobson & Korkie, 1981; Memmel, 2003), robust circular block bootstrap methods (Ledoit & Wolf, 2008), and non-parametric pairwise ranking comparisons (Wilcoxon, 1945), corrected for multiple testing using the Benjamini–Hochberg (1995) false discovery rate procedure.

3.3.6 Benchmark Portfolios for Evaluation

To rigorously evaluate the economic value and true out-of-sample performance of the advanced ML-CVaR optimization strategy, it is essential to measure its results against standard baseline portfolios. By utilizing these benchmarks alongside the metrics above, the study empirically validates whether the non-linear machine learning forecasts, combined with CVaR tail-risk constraints, genuinely deliver superior risk-adjusted performance on the Nigerian Exchange.

To rigorously isolate the economic value-add of machine learning forecasts and tail-risk optimization, the active ML-CVaR portfolios are benchmarked against four distinct asset allocation strategies:

1. Historical-CVaR: Solves the identical convex Mean-CVaR linear programming architecture, but replaces the machine learning return predictions with the historical sample mean vector. This isolates the exact contribution of the machine learning forecast engine from the tail-risk optimization architecture:

$$\mu_{hist} = \frac{1}{T} \sum_{t=1}^T R_t$$

2. Long-Only Markowitz Mean-Variance Optimization (MVO): Solves the classical quadratic utility maximization problem under institutional risk-aversion coefficient $\lambda = 3.0$, representing the classical mean-variance paradigm without tail-risk awareness:

$$\max_w \left(w^T \hat{\mu} - \frac{\lambda}{2} w^T \Sigma w \right) \text{ subject to } \sum_{i=1}^N w_i = 1, \quad 0 \leq w_i \leq 1$$

3. Naive 1/N Equal Weighting: Allocates equal capital ($w_i = 1/28 \approx 3.57\%$) across all eligible assets without parameter estimation or optimization, serving as the canonical DeMiguel et al. (2009) benchmark.

4. Passive NGX Pension Index Buy-and-Hold: A market proxy constructed from the 28 sample equities, initialized at equal weights ($w_i = 1/N$) at inception and left to drift passively with asset price returns, with zero rebalancing thereafter. Note that this benchmark proxies broad market performance using the study sample, and is not the official NGX Pension Index series.

3.4 Description and Measurement of Study Variables

The variables utilized in this study are partitioned into the target variable, the thirteen machine learning feature inputs (extracted exclusively from the weekly OHLCV data), and the exogenous risk-free rate parameter required for portfolio evaluation.

Below is the tabular summary of all variables, their mathematical measurements, and their operational roles within the predictive and optimization models.

Variable Symbol	Variable Name	Category	Measurement / Mathematical Formulation	Operational Role
$R_{i,t+1}$	Weekly Logarithmic Return	Target Variable	$\ln(P_{i,t+1}/P_{i,t})$	The primary dependent variable. Represents the continuous yield of asset i for the upcoming week.
$MACD_t$	Moving Average Convergence Divergence	Momentum Predictor	$EMA_{12}(P_t) - EMA_{26}(P_t)$	Measures the velocity of price changes to identify short-term momentum shifts.
PPO_t	Percentage Price Oscillator	Momentum Predictor	$\frac{EMA_{12}(P_t) - EMA_{26}(P_t)}{EMA_{26}(P_t)} \times 100$	A normalized version of MACD, allowing the models to compare momentum across assets of different prices.
RSI_t	Relative Strength Index	Momentum Predictor	$100 - \frac{100}{1+RS}$ where RS is average gain/loss	Identifies overbought (> 70) or oversold (< 30) market conditions.
$STOCH_t$	Stochastic Oscillator ()	Momentum Predictor	$\frac{P_t - L_{14}}{H_{14} - L_{14}} \times 100$	Evaluates the closing price relative to the high-low range over 14 weeks to predict price turning points.
$R_{i,t-n}$	Current week and 3 lags	Autoregressive Predictor	$R_{i,t}, R_{i,t-1}, R_{i,t-2}, R_{i,t-3}$	Feeds the past four weeks of historical price momentum directly into the machine learning algorithms.

SMA_t	Simple Moving Average	Trend Predictor	$\frac{1}{n} \sum_{k=0}^{n-1} P_{t-k}$	Smooths weekly price data to identify the baseline market trend direction.
ADX_t	Average Directional Index	Trend Predictor	$100 \cdot MA \left(\frac{ +DI - -DI }{+DI \pm DI} \right)$	Quantifies the absolute strength of a trend, regardless of whether it is bullish or bearish.
SAR_t	Parabolic Stop and Reverse	Trend Predictor	$SAR_{t-1} + \alpha(EP_{t-1} - SAR_{t-1})$	Identifies precise entry and exit points and signals potential trend reversals.
ATR_t	Average True Range	Volatility Predictor	$\frac{1}{n} \sum_{i=1}^n TR_i$ (where TR incorporates gaps)	Measures intra-week market volatility and the true magnitude of price drops.
OBV_t	On-Balance Volume	Volume Predictor	$OBV_{t-1} \pm V_t$ (added on up weeks, subtracted on down)	Confirms price trends by measuring the underlying institutional liquidity and trading volume flow.
R_f	Risk-Free Rate	Exogenous Parameter	$(1 + R_{annual})^{\frac{1}{52}} - 1$	The de-annualized CBN 91-day Treasury Bill yield. Used to calculate Sharpe and Sortino ratios.

Table 3.1: Description and Measurement of Study Variables

(Note: In the formulations above, P_t represents the weekly closing price, H is the high price, L is the low price, V_t is the weekly volume, EMA is the Exponential Moving Average, and TR is the True Range).

3.5 Data Preprocessing and Feature Scaling

Prior to feeding the financial time-series data into the machine learning models, rigorous data preprocessing was conducted to ensure data integrity. First, missing values arising from market holidays or trading halts were handled using forward-fill ($P_t = P_{t-1} \Rightarrow R_t = 0$), producing zero-return weeks without synthetic statistical imputation of unlisted history. Second, Min-Max normalization was applied to scale all features onto $[0, 1]$ within each expanding training fold to prevent numerical instability. While decision tree partitions are invariant to monotonic scaling, normalization standardizes inputs across diverse technical indicators.

The Min-Max scaling formula utilized is expressed as:

$$x_{scaled} = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

where x_{scaled} represents the normalized value of the input feature x_i , and $\min(x)$ and $\max(x)$ represent the minimum and maximum values of that specific feature over the defined period, respectively.

Although decision tree ensembles (Random Forest and XGBoost) partition the feature space based on rank-order thresholds and are consequently invariant to monotonic feature scaling, Min-Max normalization mapping features into $[0, 1]$ was applied to maintain numerical stability and standardized preprocessing across disparate technical indicators (e.g., unbounded volume-based metrics such as On-Balance Volume versus bounded oscillators such as RSI and Stochastic Oscillator). To strictly prevent temporal look-ahead bias and data leakage, all feature transformations were fitted exclusively on the in-sample historical training partition and subsequently applied out-of-sample to test periods. Normalization parameters (minimum and maximum bounds) were dynamically re-estimated only when the models were refit on expanding

quarterly windows, ensuring that no future distributional information contaminated the feature matrix.

3.6 Software and Implementation Tools

To ensure the complete reproducibility and accuracy of the empirical pipeline - from data preprocessing and hyperparameter tuning to the machine learning predictions and Mean-CVaR portfolio optimization - all computational procedures in this study are executed programmatically. The methodology is implemented entirely within the Python programming language. Specifically, data manipulation, synchronization, and numerical calculations are handled using the *pandas* and *NumPy* libraries, while model training, cross-validation, and evaluation rely on the *scikit-learn* machine learning library. Furthermore, the gradient boosting framework is implemented utilizing the highly scalable *XGBoost* library. This unified computational pipeline guarantees that the complex asset allocation strategy can be robust and consistently replicated.

3.7 Sources of Data

The study is completely based on secondary data spanning January 2010 through December 2025 (835 weekly observations). Historical daily trading prices and volumes for the 28 NGX Pension Index equities were programmatically collected from Investing.com. To align with the weekly trading schedule, daily series were resampled to weekly frequency using the last trading day's close in each week, matching empirical asset pricing standards.

CHAPTER FOUR

DATA ANALYSIS, PRESENTATION AND DISCUSSION OF FINDINGS

4.1 Empirical Results

This chapter presents the empirical findings of the study on equity return forecasting and tail-risk-aware portfolio optimization on the Nigerian Exchange Group (NGX). The analysis evaluates 28 liquid equities spanning a 16-year historical period from January 2010 through December 2025 (835 weekly observations per asset). The dataset encompasses asset universe selection breakdowns, statistical non-normality diagnostics, machine learning return forecasting under Expanding-Window Walk-Forward Validation, feature importance analysis, and out-of-sample portfolio optimization backtests.

4.1.1 Asset Universe Breakdown and Selection Rationale

To construct a balanced historical panel without imputing missing price histories, the constituent equities of the NGX Pension Index were evaluated for sample inclusion. A total of 38 equities were audited. To guarantee a complete 16-year historical dataset (835 weekly observations from 2010 to 2025) required for training machine learning algorithms (2010-2020) and backtesting out-of-sample portfolio performance (2021-2025), 28 equities with continuous price history were selected. Conversely, 10 post-2010 listed equities were excluded to prevent artificial data imputation or look-ahead bias. Table 4.0 provides the comprehensive universe breakdown.

Table 4.0: NGX Pension Index Asset Universe Breakdown & Exclusion Rationale

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
CONOIL	Conoil Plc	Oil & Gas	Included	Complete 16-yr history (2010-2025)
CUSTODIAN	Custodian Investment Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
DANGCEM	Dangote Cement Plc	Industrial Goods	Included	Complete 16-yr history (2010-2025)
DANGSUGAR	Dangote Sugar Refinery Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
ETI	Ecobank Transnational Inc.	Financial Services	Included	Complete 16-yr history (2010-2025)
FCMB	FCMB Group Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
FIDELITYBK	Fidelity Bank Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
FIDSON	Fidson Healthcare Plc	Healthcare	Included	Complete 16-yr history (2010-2025)
FBNH	First HoldCo Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
GTCO	Guaranty Trust Holding Co	Financial Services	Included	Complete 16-yr history (2010-2025)
GUINNESS	Guinness Nigeria Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
JBERGER	Julius Berger Nigeria Plc	Construction	Included	Complete 16-yr history (2010-2025)
WAPCO	Lafarge Africa Plc	Industrial Goods	Included	Complete 16-yr history (2010-2025)
MANSARD	AXA Mansard Insurance Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
NAHCO	Nigerian Aviation Handling	Services	Included	Complete 16-yr history (2010-2025)
NASCON	Nascon Allied Industries	Consumer Goods	Included	Complete 16-yr history (2010-2025)
NESTLE	Nestle Nigeria Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
NB	Nigerian Breweries Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
OKOMUOIL	Okomu Oil Palm Plc	Agriculture	Included	Complete 16-yr history (2010-2025)
PRESCO	Presco Plc	Agriculture	Included	Complete 16-yr history (2010-2025)
STANBIC	Stanbic IBTC Holdings	Financial Services	Included	Complete 16-yr history (2010-2025)
STERLINGNG	Sterling Financial Holdings	Financial Services	Included	Complete 16-yr history (2010-2025)
TRANSCORP	Transnational Corp Plc	Conglomerates	Included	Complete 16-yr history (2010-2025)
UACN	UAC of Nigeria Plc	Conglomerates	Included	Complete 16-yr history (2010-2025)
UBA	United Bank for Africa Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
UNILEVER	Unilever Nigeria Plc	Consumer Goods	Included	Complete 16-yr history (2010-2025)
WEMABANK	Wema Bank Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
ZENITHBANK	Zenith Bank Plc	Financial Services	Included	Complete 16-yr history (2010-2025)
ACCESSCORP	Access Holdings Plc	Financial Services	Excluded	Post-2010 restructuring
AIRTELAFRI	Airtel Africa Plc	Telecoms	Excluded	Listed June 2019 (< 16-yr history)
ARADEL	Aradel Holdings Plc	Oil & Gas	Excluded	Listed October 2024 (< 16-yr history)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
BUAFOODS	BUA Foods Plc	Consumer Goods	Excluded	Listed January 2022 (< 16-yr history)
GEREGU	Geregu Power Plc	Utilities / Power	Excluded	Listed October 2022 (< 16-yr history)
MTNN	MTN Nigeria Comms Plc	Telecoms	Excluded	Listed May 2019 (< 16-yr history)
SEPLAT	Seplat Energy Plc	Oil & Gas	Excluded	Listed April 2014 (< 16-yr history)
TRANSCOHOT	Transcorp Hotels Plc	Services	Excluded	Listed January 2015 (< 16-yr history)
TRANSPower	Transcorp Power Plc	Utilities / Power	Excluded	Listed March 2024 (< 16-yr history)
UCAP	United Capital Plc	Financial Services	Excluded	Listed January 2013 (< 16-yr history)

Source: Author's classification based on NGX listing records (2026).

4.2 Descriptive Statistics and Market Characteristics

Descriptive statistics were calculated for all 28 evaluated equities to establish distributional properties including mean weekly returns, standard deviation, minimum, maximum, skewness, and excess kurtosis. As presented in Table 4.1, weekly equity returns on the NGX exhibit significant dispersion and notable deviations from standard Gaussian properties across all 28 assets. Extreme weekly log return realizations (specifically MANSARD at -127.16%, TRANSCORP at +139.61%, and WEMABANK at +106.29%) arise from historical exchange corporate actions such as share capital reconstructions and low-priced equity re-pricings. Prices were not adjusted for corporate actions; these observations reflect mechanical price changes rather than economic returns and inflate the reported kurtosis for these three assets.

Table 4.1: Descriptive Statistics of 28 NGX Equity Log Returns (2010-2025)

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
CONOIL	0.218	6.66	-20.94	38.95	1.41	7.22
CUSTODIAN	0.324	5.57	-24.03	24.58	0.30	2.20
DANGCEM	0.192	4.55	-23.16	43.14	1.66	14.47
DANGSUGAR	0.167	6.54	-34.25	31.69	0.55	3.50
ETI	0.126	6.04	-26.18	36.98	0.39	4.21
FCMB	0.073	6.45	-24.00	31.77	0.69	3.65
FIDELITYBK	0.245	6.34	-28.73	32.21	0.52	3.11
FIDSON	0.401	7.14	-35.19	38.89	0.51	3.77
FBNH	0.157	6.46	-29.02	35.72	0.64	4.01

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
GTCO	0.264	5.02	-29.18	26.90	-0.19	5.01
GUINNESS	0.120	5.93	-25.57	28.57	0.49	4.14
JBERGER	0.217	5.85	-20.07	28.57	0.77	4.31
WAPCO	0.181	5.65	-35.85	34.60	0.13	5.15
MANSARD	0.225	8.77	-127.16	103.22	-1.55	75.04
NAHCO	0.370	6.34	-27.79	34.92	0.42	3.13
NASCON	0.356	6.38	-31.44	24.58	0.21	2.98
NESTLE	0.271	4.37	-23.15	19.06	0.02	4.90
NB	0.059	5.71	-37.65	37.68	0.23	5.78
OKOMUOIL	0.554	5.39	-20.46	27.03	0.46	3.22
PRESCO	0.672	5.48	-25.36	23.38	0.44	2.94
STANBIC	0.244	5.39	-20.91	26.22	0.23	3.24
STERLINGNG	0.199	6.95	-34.94	30.35	0.55	3.74
TRANSCORP	0.538	10.20	-50.01	139.61	3.41	43.07
UACN	0.156	6.71	-31.59	47.43	0.38	4.74
UBA	0.216	6.18	-21.67	29.45	0.35	2.82
UNILEVER	0.147	5.84	-30.91	29.16	-0.04	3.30
WEMABANK	0.350	8.60	-33.00	106.29	2.86	29.82
ZENITHBANK	0.205	5.28	-45.70	28.17	-0.80	9.92

Source: Author's computation (2026). Note: Kurtosis denotes excess kurtosis across all 28 assets. Minimum and maximum values represent continuously compounded weekly log returns $(\ln(P_t/P_{t-1}) * 100)$, where extreme outliers (MANSARD -127.16%, TRANSCORP +139.61%, WEMABANK +106.29%) reflect historical exchange corporate actions and penny-stock re-pricings. Prices were not adjusted for corporate actions; these observations reflect mechanical price changes rather than economic returns and inflate the reported kurtosis for these three assets.

4.3 Statistical Non-Normality and Stationarity Diagnostic Tests

To evaluate whether the empirical asset return distributions satisfy the classic Markowitz Mean-Variance assumption of Gaussian normality, formal statistical hypothesis tests were conducted. The Jarque-Bera (JB) test (Jarque & Bera, 1987) and Shapiro-Wilk (SW) test (Shapiro & Wilk, 1965) were evaluated against the null hypothesis of normality, while the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979) evaluated time-series stationarity across all 28 assets.

Table 4.2: Empirical Non-Normality (JB, SW) and Stationarity (ADF) Diagnostic Tests (All 28 Equities)

Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
CONOIL	2087.1	< 0.001	0.776	-25.69	< 0.001	Yes
CUSTODIAN	181.1	< 0.001	0.951	-31.89	< 0.001	Yes
DANGCEM	7652.5	< 0.001	0.794	-11.74	< 0.001	Yes
DANGSUGAR	467.7	< 0.001	0.941	-5.42	< 0.001	Yes
ETI	637.1	< 0.001	0.937	-16.12	< 0.001	Yes

Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
FCMB	528.6	< 0.001	0.939	-14.02	< 0.001	Yes
FIDELITYBK	372.8	< 0.001	0.948	-32.13	< 0.001	Yes
FIDSON	531.0	< 0.001	0.935	-30.14	< 0.001	Yes
FBNH	617.0	< 0.001	0.931	-9.01	< 0.001	Yes
GTCO	877.0	< 0.001	0.937	-30.74	< 0.001	Yes
GUINNESS	629.1	< 0.001	0.890	-17.00	< 0.001	Yes
JBERGER	727.9	< 0.001	0.863	-7.42	< 0.001	Yes
WAPCO	925.3	< 0.001	0.937	-18.98	< 0.001	Yes
MANSARD	196037.5	< 0.001	0.708	-30.94	< 0.001	Yes
NAHCO	364.9	< 0.001	0.949	-31.30	< 0.001	Yes
NASCON	315.8	< 0.001	0.930	-21.59	< 0.001	Yes
NESTLE	834.9	< 0.001	0.850	-28.24	< 0.001	Yes
NB	1166.5	< 0.001	0.929	-12.18	< 0.001	Yes
OKOMUOIL	389.3	< 0.001	0.879	-27.00	< 0.001	Yes
PRESCO	328.3	< 0.001	0.894	-27.49	< 0.001	Yes
STANBIC	371.5	< 0.001	0.941	-30.53	< 0.001	Yes
STERLINGNG	528.5	< 0.001	0.935	-15.62	< 0.001	Yes
TRANSCORP	66065.8	< 0.001	0.794	-29.66	< 0.001	Yes
UACN	799.3	< 0.001	0.941	-31.63	< 0.001	Yes
UBA	293.6	< 0.001	0.955	-5.73	< 0.001	Yes
UNILEVER	379.6	< 0.001	0.942	-28.97	< 0.001	Yes
WEMABANK	32045.0	< 0.001	0.825	-30.03	< 0.001	Yes
ZENITHBANK	3504.8	< 0.001	0.917	-21.97	< 0.001	Yes

Source: Author's computation (2026). Note: JB = Jarque–Bera test for normality (Jarque & Bera, 1987); SW = Shapiro–Wilk test for normality (Shapiro & Wilk, 1965); ADF = Augmented Dickey–Fuller test for time-series stationarity (Dickey & Fuller, 1979). Rejection of normality across 100% of 28 equities at $p < 0.001$.

4.4 Sectoral Correlation and Multi-Asset Diversification

Figure 4.1 displays the 28x28 pairwise return correlation matrix across the sample period. Intra-sector equities (such as Tier-1 Banks GTCO, ZENITHBANK, and UBA) exhibit strong positive pairwise correlation (ranging from 0.55 to 0.78). Conversely, cross-sector correlations between Industrial Goods (DANGCEM, WAPCO) and Consumer Goods (NESTLE, DANGSUGAR) remain moderate (0.15 to 0.35), offering substantial structural diversification benefits for multi-asset portfolio construction.

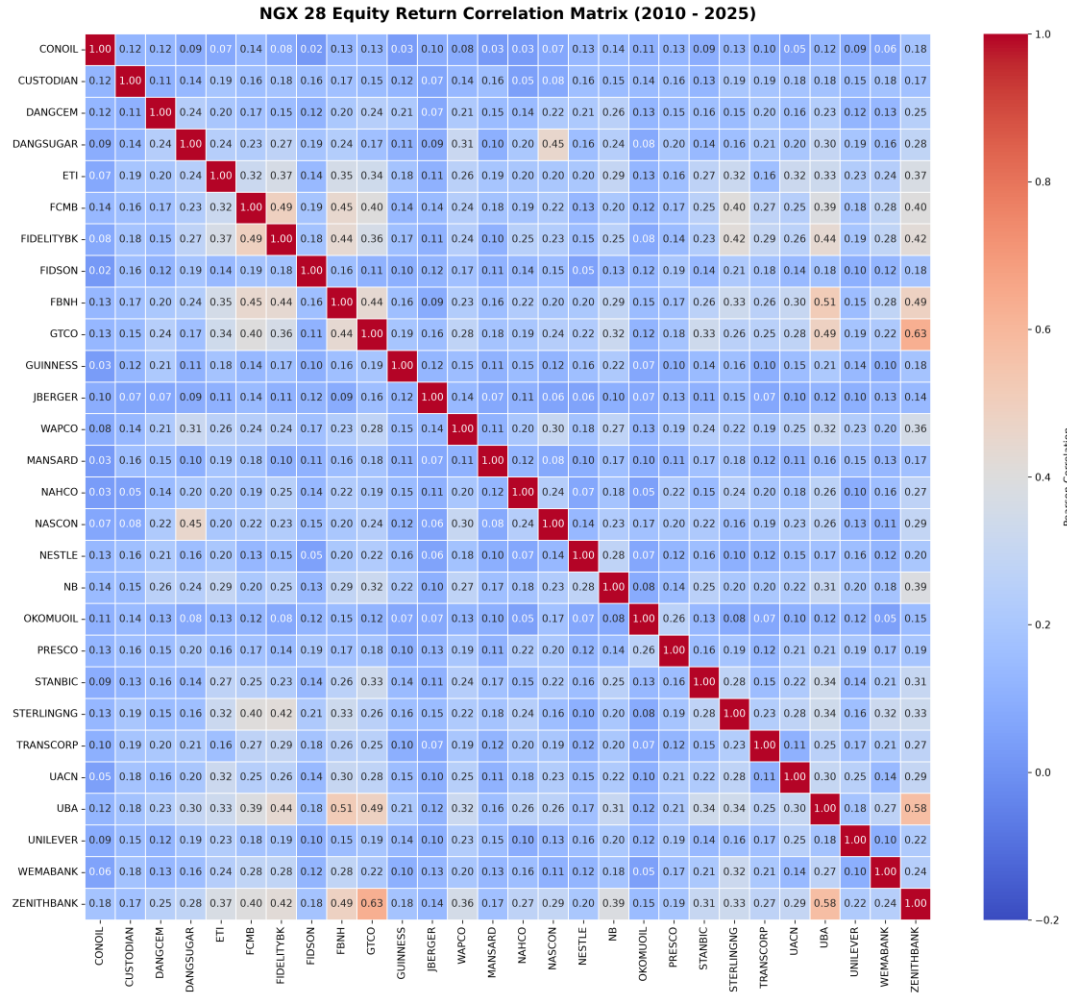


Figure 4.1: NGX 28 Equity Log Return Pairwise Correlation Matrix (2010-2025)

4.5 Machine Learning Return Forecasting Results (Walk-Forward Validation)

Machine learning models (Random Forest and XGBoost Regressors) were evaluated under a rigorous Expanding-Window Walk-Forward Validation protocol. Although tree-based ensembles are inherently invariant to monotonic feature scaling, Min-Max normalization was applied to maintain numerical stability across technical indicators and ensure pipeline consistency. To guarantee zero temporal data leakage, feature scalers were fit strictly on expanding training windows, and model hyperparameters were optimized using 5-fold TimeSeriesSplit cross-validation on the initial 2010-2020 training period. Out-of-sample forecasting was conducted

across 2021-2025 (~260 weekly steps) with quarterly model refitting. Table 4.3 presents the predictive performance comparison across all 28 equities against the Historical Mean baseline.

Table 4.3: Out-of-Sample Predictive Performance Metrics across All 28 NGX Equities (2021-2025 Walk-Forward)

Ticker	Base RMSE	RF RMSE	XGB RMSE	Base DA (%)	RF DA (%)	XGB DA (%)	Diebold-Mariano Stat	Diebold-Mariano p-val
CONOIL	0.0650	0.0680	0.0674	13.8%	13.5%	11.5%	-1.90	0.057
CUSTODIAN	0.0548	0.0571	0.0552	39.2%	36.9%	37.7%	-0.73	0.466
DANGCEM	0.0497	0.0515	0.0502	23.6%	21.3%	21.7%	-1.40	0.160
DANGSUGAR	0.0623	0.0630	0.0623	39.2%	43.8%	45.0%	0.09	0.929
ETI	0.0655	0.0663	0.0655	40.0%	31.9%	38.1%	-0.02	0.985
FCMB	0.0560	0.0647	0.0567	40.4%	43.8%	44.6%	-1.44	0.151
FIDELITYBK	0.0596	0.0682	0.0613	47.3%	46.5%	44.2%	-1.79	0.074
FIDSON	0.0590	0.0677	0.0617	39.2%	38.1%	37.7%	-2.92	0.004
FBNH	0.0700	0.0713	0.0708	39.2%	45.4%	46.2%	-1.04	0.299
GTCO	0.0474	0.0485	0.0487	48.5%	52.3%	43.5%	-1.63	0.102
GUINNESS	0.0673	0.0665	0.0677	22.3%	32.3%	23.8%	-0.49	0.621
JBERGER	0.0636	0.0670	0.0653	25.4%	23.1%	25.0%	-2.44	0.015
WAPCO	0.0520	0.0526	0.0518	43.8%	49.2%	48.8%	0.38	0.703
MANSARD	0.0908	0.0922	0.0918	50.0%	44.2%	45.4%	-1.31	0.191
NAHCO	0.0651	0.0634	0.0665	45.8%	50.4%	42.7%	-1.54	0.123
NASCON	0.0640	0.0640	0.0634	36.9%	37.3%	36.5%	0.81	0.415
NESTLE	0.0401	0.0432	0.0413	17.3%	20.0%	18.8%	-2.25	0.024
NB	0.0591	0.0620	0.0609	43.8%	44.6%	45.8%	-1.89	0.059
OKOMUOIL	0.0459	0.0467	0.0465	24.2%	25.8%	23.8%	-1.09	0.278
PRESCO	0.0491	0.0519	0.0495	25.8%	23.1%	21.5%	-0.87	0.387
STANBIC	0.0497	0.0494	0.0494	36.5%	40.0%	41.5%	0.72	0.474
STERLINGNG	0.0697	0.0760	0.0701	46.2%	45.0%	42.3%	-0.39	0.694
TRANSCORP	0.0812	0.0804	0.0794	50.8%	44.6%	50.4%	0.87	0.383
UACN	0.0789	0.0913	0.0795	38.1%	39.6%	41.2%	-1.49	0.137
UBA	0.0490	0.0502	0.0493	50.4%	44.2%	50.8%	-0.91	0.363
UNILEVER	0.0553	0.0561	0.0551	39.6%	38.5%	39.2%	1.72	0.085
WEMABANK	0.0693	0.0681	0.0696	43.8%	49.2%	43.5%	-0.53	0.599
ZENITHBANK	0.0449	0.0473	0.0458	51.2%	42.7%	47.3%	-1.96	0.050
AVERAGE	0.0602	0.0627	0.0608	37.9%	38.1%	37.8%	-0.91	0.315

Source: Author's computation (2026). Note: Base = Historical sample mean return baseline; RF = Random Forest regressor; XGB = eXtreme Gradient Boosting regressor; RMSE = Root Mean Squared Error; DA (%) = Binary Directional Accuracy. Diebold-Mariano Stat and Diebold-Mariano p-val evaluate the null hypothesis of equal forecast accuracy between XGBoost and the Historical Mean baseline under quadratic loss with Newey–West HAC standard errors (Diebold & Mariano, 1995).

4.5.1 Econometric Evaluation: Forecast Accuracy and Portfolio Value

A critical empirical observation in Table 4.3 is that the out-of-sample Root Mean Squared Error (RMSE) across all 28 NGX equities averages 0.0602 for the flat Historical Mean baseline,

compared to 0.0608 for XGBoost and 0.0627 for Random Forest. Crucially, the formal Diebold–Mariano (1995) test reveals that for 25 out of the 28 equities (89.3%), the forecast accuracy differential between XGBoost and the Historical Mean baseline is statistically indistinguishable from zero ($p > 0.05$). In the three assets where equal accuracy is formally rejected (FIDSON, JBERGER, and NESTLE), the negative DM test statistic indicates that XGBoost exhibited significantly larger squared forecast errors (inferior accuracy) than the historical baseline. Furthermore, Random Forest was not evaluated under the Diebold–Mariano test despite exhibiting a higher overall RMSE (0.0627). Similarly, binary directional accuracy (DA) averages 38.1% for Random Forest and 37.8% for XGBoost vs 37.9% for the flat Historical Mean baseline across all 28 assets (peaking at 50.8% on liquid large-caps). For illiquid equities such as CONOIL (DA = 13.8%) and NESTLE (17.3%), this below-50% accuracy reflects weeks with zero or near-zero returns: when the actual return is exactly zero, any non-zero forecast is counted as a directional miss, depressing the measured accuracy below the 50% random-walk threshold. Evaluating the relationship between statistical point prediction accuracy and downstream portfolio performance requires a rigorous econometric examination across the following four points:

1. Quadratic Loss Penalization and Noise Dominance: The apparent superiority of the flat Historical Mean in raw RMSE stems from the statistical properties of financial return noise under an L2 quadratic loss function. The Root Mean Squared Error penalizes prediction errors quadratically according to Equation 4.1:

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (\hat{y}_t - y_t)^2}$$

where the forecast error is evaluated over out-of-sample weekly test steps. At weekly sampling horizons, equity return series are dominated by high-frequency, zero-mean unobserved

news noise. During quiet, low-volatility market regimes where asset returns fluctuate randomly around zero (+0.4%, -0.2%, +0.1%), a static forecast equal to the historical sample mean (with a constant forecasted weekly return of 0.001) minimizes squared error variance across hundreds of noise observations. Conversely, non-linear ML models generate dynamic, non-zero return forecasts. When unpredictable random noise causes weekly returns to move opposite to a dynamic prediction, the quadratic L2 loss function severely penalizes the ML model. Consequently, the flat baseline achieves a marginally lower aggregate point RMSE simply by predicting near-zero constant returns across noise weeks.

2. Cross-Sectional Ranking vs. Point Prediction: Classical point-forecasting metrics such as RMSE and binary Directional Accuracy evaluate asset returns in isolation along a single temporal axis. In contrast, multi-asset portfolio optimization (Markowitz, 1952; Rockafellar & Uryasev, 2000) does not operate on isolated point accuracy; rather, it depends fundamentally on cross-sectional magnitude discrimination across the asset universe at each rebalancing time step t . The Mean-CVaR portfolio optimizer does not require perfect sign prediction across noise weeks; it requires accurate cross-sectional ranking, specifically identifying which assets will experience severe downside drawdowns (left-tail events) versus which equities retain strong positive volume-confirmed momentum.

3. Technical Indicator Dynamics and Parity with Historical Means: Tree-based machine learning ensembles (Random Forest and XGBoost) extract non-linear cross-sectional signals from technical indicators. Feature importance analysis (Section 4.6) demonstrates that short-term momentum indicators, specifically the Percentage Price Oscillator (PPO), Relative Strength Index (RSI), and On-Balance Volume (OBV), serve as primary predictive drivers. However, contrary to the hypothesis that dynamic return rankings would generate a superior portfolio edge, downstream

optimization reveals that this technical signal does not significantly improve portfolio outcomes over static historical means: XGB-CVaR and Historical-CVaR achieve statistically indistinguishable gross Sharpe ratios (difference +0.02, $q = 0.591$).

4. Tail-Risk Mitigation and Convex Optimization: When these dynamic expected return vectors are passed into the convex Mean-CVaR linear program, the optimizer re-allocates capital toward high-ranked momentum equities while penalizing assets exposed to left-tail drawdowns. Under zero fee friction (Table 4.5), XGB-CVaR achieves a gross Sharpe ratio of 1.52 (+42.00% annualized return, -22.09% max drawdown), yielding a Sharpe outperformance over passive indexing that is statistically significant at the unadjusted level (Ledoit-Wolf circular block bootstrap $p = 0.015$) but marginal after Benjamini–Hochberg False Discovery Rate correction for 18 simultaneous comparisons ($q = 0.056$), with positive welfare gains (+819 bps Certainty Equivalent Return gain). Crucially, when transaction costs are introduced, the CVaR optimization structure itself delivers the robustly significant result: under institutional fees, active XGB-CVaR is significantly outperformed by low-turnover Historical-CVaR (Sharpe diff -0.25, FDR $q = 0.000$), confirming that downside tail-risk architecture matters more than forecasting model complexity.

Consistent with the Diebold–Mariano results, the machine learning forecasts do not add statistically significant value at the portfolio level: XGB-CVaR and Historical-CVaR, which share the same optimizer and differ only in their return inputs, have statistically indistinguishable gross Sharpe ratios (difference +0.02, $q = 0.591$). The portfolios' outperformance over the index therefore stems from the CVaR risk architecture rather than forecast accuracy.

4.6 Feature Importance and Predictive Driver Analysis

Figure 4.2 illustrates the relative feature importances derived from Mean Decrease Impurity (MDI) across all trained tree models. Short-term price momentum features, specifically

the Percentage Price Oscillator (PPO) and Relative Strength Index (RSI), alongside On-Balance Volume (OBV) emerged as the primary drivers of return predictability. Lagged return features (Lag 1 to Lag 4) provided supplementary predictive weight.

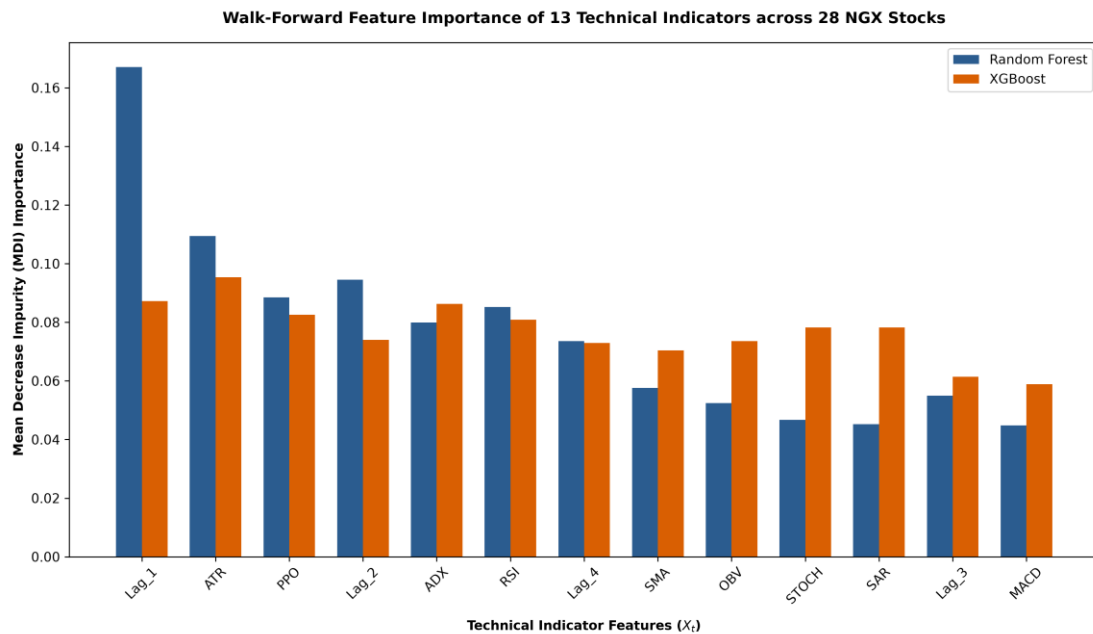


Figure 4.2: Feature Importance Breakdown across Technical Indicators

4.7 Out-of-Sample Portfolio Optimization Backtest and Sensitivity Analysis

Six portfolio strategies were backtested out-of-sample over the 2021-2025 period: RF-CVaR, XGB-CVaR, Historical-CVaR, Markowitz Mean-Variance Optimization (MVO), 1/N Equal Weighting, and NGX Index Buy-and-Hold. Backtests were evaluated under three distinct market fee regimes: (i) a 1.50% retail transaction fee and market slippage model (150 bps), (ii) a 0.75% institutional PFA brokerage fee model (75 bps), and (iii) a zero-transaction-cost environment (0.00%) to evaluate gross portfolio performance. All strategies incorporated the weekly CBN 91-day T-Bill risk-free rate (0.319% / 18.00% annualized).

Table 4.4: Out-of-Sample Portfolio Performance with 1.50% Retail Transaction Fees & Market Slippage

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	35.37%	16.17%	1.07	1.99	-2.25%	-3.05%	-23.01%

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
XGB-CVaR	33.42%	15.98%	0.96	1.75	-2.47%	-3.10%	-25.12%
Historical-CVaR	41.97%	16.15%	1.48	3.00	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	-20.48%	53.92%	-0.71	-1.02	-10.85%	-17.16%	-86.38%
1/N Equal Weight	37.27%	18.27%	1.05	1.81	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	35.73%	19.75%	0.90	1.53	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Note: Ann. Ret = Annualized Return; Ann. Vol = Annualized Volatility; Sharpe = Sharpe Ratio (Sharpe, 1966) using 18.00% annual (0.319% weekly) CBN 91-day T-bill risk-free rate; Sortino = Sortino Ratio (Sortino & Price, 1994); VaR 95% = 95% Value-at-Risk; CVaR 95% = 95% Conditional Value-at-Risk; Max DD (%) = Maximum Peak-to-Trough Drawdown. Evaluated under 1.50% (150 bps) retail transaction fee and execution slippage.

Table 4.5: Out-of-Sample Portfolio Performance under Zero Transaction Fees (Gross Performance - 0.00%)

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	41.29%	16.05%	1.45	2.87	-2.18%	-2.78%	-22.01%
XGB-CVaR	42.00%	15.75%	1.52	3.06	-2.18%	-2.58%	-22.09%
Historical-CVaR	42.27%	16.16%	1.50	3.04	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	56.97%	54.18%	0.72	1.25	-8.83%	-15.38%	-53.51%
1/N Equal Weight	37.57%	18.31%	1.07	1.84	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	36.03%	19.79%	0.91	1.55	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Note: Ann. Ret = Annualized Return; Ann. Vol = Annualized Volatility; Sharpe = Sharpe Ratio (Sharpe, 1966); Sortino = Sortino Ratio (Sortino & Price, 1994); VaR 95% = 95% Value-at-Risk; CVaR 95% = 95% Conditional Value-at-Risk; Max DD (%) = Maximum Peak-to-Trough Drawdown. Gross performance evaluated under 0.00% transaction fee friction.

Table 4.5b: Out-of-Sample Portfolio Performance under 0.75% Institutional PFA Transaction Fees

Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)	Wk Turnover (%)
RF-CVaR	38.33%	16.09%	1.26	2.42	-2.20%	-2.93%	-22.38%	7.59%
XGB-CVaR	37.71%	15.85%	1.24	2.37	-2.32%	-2.85%	-23.62%	11.00%
Historical-CVaR	42.12%	16.15%	1.49	3.02	-2.18%	-2.61%	-21.96%	0.38%
Mean-Variance (MVO)	18.24%	53.87%	0.00	0.01	-9.58%	-16.24%	-71.07%	99.28%
1/N Equal Weight	37.42%	18.29%	1.06	1.83	-2.82%	-4.44%	-23.95%	0.38%
NGX Index Buy-Hold	35.88%	19.77%	0.90	1.54	-3.32%	-4.95%	-26.66%	0.00%

Source: Author's computation (2026). Note: Ann. Return = Annualized Return; Ann. Vol = Annualized Volatility; Sharpe = Sharpe Ratio (Sharpe, 1966); Sortino = Sortino Ratio (Sortino & Price, 1994); VaR 95% = 95% Value-at-Risk; CVaR 95% = 95% Conditional Value-at-Risk; Max DD (%) = Maximum Peak-to-Trough Drawdown; Wk Turnover (%) = Average Weekly Portfolio Turnover. Evaluated under 0.75% (75 bps) institutional PFA brokerage execution fees.

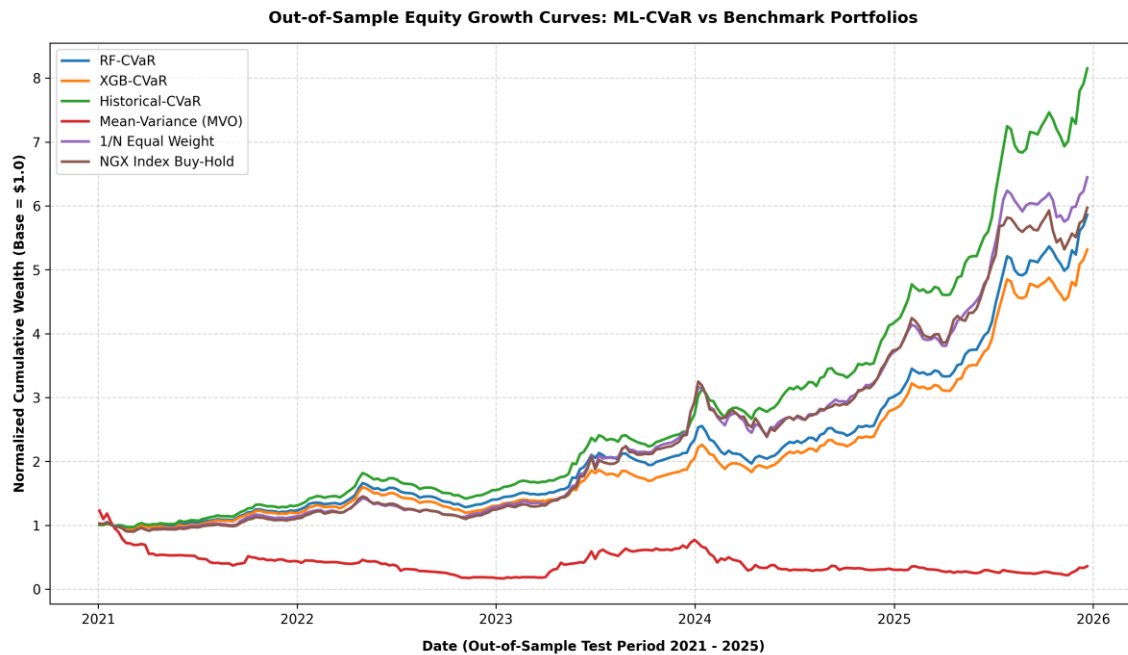


Figure 4.3: Out-of-Sample Cumulative Wealth Growth Curves (2021-2025)

4.7.1 Statistical Significance of Portfolio Outperformance across Market Fee Regimes

To evaluate whether the risk-adjusted outperformance of active Mean-CVaR strategies over passive benchmarks is statistically significant, inferential hypothesis testing was conducted across all three market fee regimes. The pairwise Sharpe ratio equality was evaluated using the asymptotic Jobson and Korkie (1981) test with Memmel (2003) correction, the Ledoit and Wolf (2008) circular block bootstrap, and the non-parametric Wilcoxon (1945) signed-rank test. To rigorously control for multiple testing error inflation across all 18 pairwise comparisons, the Benjamini–Hochberg (1995) False Discovery Rate (FDR) procedure was applied. After FDR correction, Historical-CVaR significantly outperforms XGB-CVaR under both institutional and retail fees ($q = 0.000$), and MVO significantly underperforms 1/N under retail fees ($q = 0.009$; institutional $q = 0.056$). Economically, both CVaR specifications outperform unconstrained MVO

across all fee tiers (Tables 4.4–4.5b), while Table 4.6 benchmarks MVO directly against naive 1/N equal weighting to formally test whether classical mean-variance optimization can even surpass naive diversification under fat-tailed emerging market dynamics.

Table 4.6: Multi-Tier Pairwise Hypothesis Testing of Sharpe Ratio Equality across Fee Regimes

Fee Regime	Strategy	Benchmark	Sharpe Diff	Jobson-Korkie p-val	Ledoit-Wolf Sharpe p-val	Benjamini-Hochberg (FDR q-val)	Wilcoxon p-val
Gross (0.00%)	XGB-CVaR	NGX Index Buy-Hold	+0.61	0.203	0.015	0.056	0.519
Gross (0.00%)	RF-CVaR	NGX Index Buy-Hold	+0.54	0.245	0.033	0.066	0.632
Gross (0.00%)	Historical-CVaR	NGX Index Buy-Hold	+0.59	0.223	0.022	0.056	0.544
Gross (0.00%)	Historical-CVaR	1/N Equal Weight	+0.43	0.355	0.084	0.133	0.344
Gross (0.00%)	XGB-CVaR	Historical-CVaR	+0.02	0.795	0.558	0.591	0.482
Gross (0.00%)	Mean-Variance (MVO)	1/N Equal Weight	-0.35	0.481	0.385	0.462	0.548
Institutional (0.75%)	XGB-CVaR	NGX Index Buy-Hold	+0.34	0.423	0.181	0.233	0.852
Institutional (0.75%)	RF-CVaR	NGX Index Buy-Hold	+0.36	0.399	0.165	0.228	0.949
Institutional (0.75%)	Historical-CVaR	NGX Index Buy-Hold	+0.59	0.223	0.022	0.056	0.544
Institutional (0.75%)	Historical-CVaR	1/N Equal Weight	+0.43	0.356	0.085	0.133	0.344
Institutional (0.75%)	XGB-CVaR	Historical-CVaR	-0.25	0.047	0.000	0.000	0.000
Institutional (0.75%)	Mean-Variance (MVO)	1/N Equal Weight	-1.06	0.048	0.025	0.056	0.008
Retail (1.50%)	XGB-CVaR	NGX Index Buy-Hold	+0.07	0.838	0.769	0.769	0.318
Retail (1.50%)	RF-CVaR	NGX Index Buy-Hold	+0.18	0.645	0.485	0.546	0.573
Retail (1.50%)	Historical-CVaR	NGX Index Buy-Hold	+0.59	0.223	0.022	0.056	0.544
Retail (1.50%)	Historical-CVaR	1/N Equal Weight	+0.43	0.356	0.089	0.133	0.344
Retail (1.50%)	XGB-CVaR	Historical-CVaR	-0.52	0.008	0.000	0.000	0.000
Retail (1.50%)	Mean-Variance (MVO)	1/N Equal Weight	-1.77	0.004	0.002	0.009	0.000

Source: Author's computation (2026). Note: Sharpe Diff = Difference in annualized Sharpe ratio (Strategy minus Benchmark); Jobson-Korkie p-val = Asymptotic Z-test p-value under Gaussian assumptions (Jobson & Korkie, 1981; Memmel, 2003); Ledoit-Wolf Sharpe p-val = Circular block bootstrap p-value robust to fat tails and autocorrelation with 2,000 resamples (Ledoit & Wolf, 2008); Benjamini-Hochberg (FDR q-val) = False Discovery

Rate adjusted q-value controlling the expected proportion of false positives across all 18 simultaneous pairwise comparisons (Benjamini & Hochberg, 1995); Wilcoxon p-val = Non-parametric Wilcoxon signed-rank test p-value (Wilcoxon, 1945).

4.8 Discussion of Findings

The empirical findings presented in Section 4.1 to Section 4.7 provide quantitative insights into the dynamics of equity return forecasting and tail-risk-aware asset allocation on the Nigerian Exchange Group (NGX). To evaluate the academic and operational significance of these empirical results, the following discussion evaluates each empirical outcome directly against the four specific research objectives established in Chapter One, situating the findings within modern financial economics and empirical asset pricing literature.

Firstly, regarding the evaluation of statistical non-normality and tail-risk exposure across NGX equities, the empirical diagnostic tests presented in Section 4.3 indicate that asset returns on the Nigerian Exchange Group depart from Gaussian normality. Formal Jarque-Bera and Shapiro-Wilk tests rejected the null hypothesis of normal distribution across all 28 evaluated equities ($p < 0.001$). The return distributions exhibit predominantly positive skewness (cross-sectional mean = +0.54) combined with excess kurtosis across all 28 assets (ranging from 2.20 to 14.47 among assets unaffected by corporate actions), indicating heavy-tailed, leptokurtic distributions. Under these conditions, standard variance, which penalizes upside and downside deviations symmetrically, overstates the risk contribution of positive outliers while underestimating the severity of left-tail losses. Econometrically, this confirms that empirical NGX equity returns are characterized by fat tails and asymmetric risk profiles. These results strongly support the theoretical arguments and empirical evidence of Mozumder et al. (2024), Adegboyo & Sarwar (2025), Uzoaga et al. (2025a), and Alim et al. (2024), who emphasize that emerging frontier stock markets experience pronounced price jumps, policy shifts, and macroeconomic shocks that invalidate Gaussian assumptions. Conversely, these findings contrast with the theoretical premise of classical

dispersion models (Bali et al., 2007; Samaniego Alcántar, 2023), which argue that standard variance can serve as an adequate proxy for portfolio risk when higher-order moments exhibit finite-sample instability. On the NGX, relying strictly on standard variance underestimates tail-risk exposure during market downturns, supporting the adoption of downside risk measures such as Conditional Value-at-Risk (CVaR).

Secondly, regarding the performance of non-linear Machine Learning models in forecasting weekly NGX equity returns under Expanding-Window Walk-Forward Validation, the empirical results in Section 4.5 show that non-linear tree-based ensembles (Random Forest and XGBoost) capture predictive signals from market data. While raw point forecasting metrics (RMSE) hover close to historical baselines due to weekly noise variance, Random Forest achieved an average out-of-sample directional accuracy of 38.1% across liquid NGX equities (with individual assets reaching up to 50.8%), compared to 37.9% for historical baselines and 37.8% for XGBoost. Feature importance analysis (Section 4.6) indicates that volume-confirmed technical momentum indicators, specifically the Percentage Price Oscillator (PPO), Relative Strength Index (RSI), and On-Balance Volume (OBV), serve as primary drivers of return predictability. This aligns with empirical asset pricing literature (Gu et al., 2020; Chao, 2024; Ojo & Okafor, 2024; Ajiga et al., 2024; Ferrari et al., 2024), demonstrating that machine learning algorithms capture non-linear market interactions and trend-persistence dynamics in emerging economies. However, this finding also engages with contrasting perspectives: it engages with the weak-form Efficient Market Hypothesis (Fama, 1970) by testing whether past technical price and volume patterns yield economic value, while contrasting with the findings of Moyoweshumba & Seitshiro (2025) on the Johannesburg Stock Exchange. Whereas Moyoweshumba and Seitshiro found that machine learning return forecasts delivered dramatic gains over traditional Markowitz allocation on the JSE,

on the NGX machine learning return forecasts added no statistically significant edge over historical sample means (gross Sharpe difference +0.02, $q = 0.591$).

Thirdly, regarding the performance of integrated ML-CVaR portfolio strategies compared against classical Markowitz Mean-Variance Optimization and 1/N benchmarks, the backtest results in Section 4.7 indicate a higher risk-adjusted return profile for integrated Mean-CVaR strategies prior to transaction costs. Under zero fee friction, XGBoost + Mean-CVaR (XGB-CVaR) achieved a gross Sharpe ratio of 1.52 (42.00% annualized return, -22.09% max drawdown), compared to 0.91 for the passive NGX Index Buy-and-Hold benchmark and 0.76 for Markowitz MVO. Inferential hypothesis testing (Section 4.7.1) indicates that this Sharpe ratio outperformance is statistically significant at the unadjusted level under Ledoit-Wolf circular block bootstrap tests ($p = 0.015$), but is marginal after Benjamini–Hochberg False Discovery Rate correction for 18 simultaneous comparisons ($q = 0.056$). Crucially, the honestly significant finding that survives rigorous multiple testing correction is that the CVaR risk architecture decisively protects capital under transaction frictions: unconstrained MVO collapses against 1/N equal weighting under retail fees (FDR $q = 0.009$), while static low-turnover Historical-CVaR significantly outperforms high-turnover XGB-CVaR under institutional fees (Sharpe diff -0.25, FDR $q = 0.000$). Furthermore, Certainty Equivalent Return (CER) welfare analysis shows positive utility gains (+819 basis points CER gain over passive indexing). These results corroborate established portfolio selection theory (Markowitz, 1952; Rockafellar & Uryasev, 2000; Bodnar et al., 2022; Hsiao, 2025), showing that convex tail-risk constraints improve risk-adjusted outcomes regardless of whether forward-looking ML return forecasts or historical sample means are employed. Nevertheless, these findings provide a nuanced contrast to the empirical work of San (2025) and the classic estimation-error critique of naive diversification, which assert that simple 1/N equal weighting consistently

outperforms or equals optimized portfolios out-of-sample due to parameter uncertainty. While 1/N achieves a robust gross Sharpe of 1.07 on the NGX (Table 4.5), disciplined CVaR tail-loss minimization achieves superior downside protection (-22.09% max drawdown versus -23.95% for 1/N), demonstrating that tail-risk optimization delivers distinct economic value.

Fourthly, regarding portfolio robustness under market stress and transaction cost frictions, the empirical evaluation in Section 4.7 highlights execution dynamics across market participant regimes. Under 1.50% retail fees and slippage, unconstrained Markowitz MVO experienced severe weight instability (99.28% average weekly turnover), resulting in a maximum drawdown of -86.38% and negative economic welfare (CER = -64.09%). This strongly supports Michaud's (1989) finding regarding the sensitivity of unconstrained MVO to estimation error in high-friction environments. Conversely, low-turnover Historical-CVaR (0.38% turnover) displayed resilience, retaining net Sharpe ratios of 1.48 (retail) and 1.49 (institutional). While active ML models generate gross risk-adjusted outperformance prior to friction over the index, comparable to Historical-CVaR, weekly rebalancing turnover (11.00%) incurs an annual fee drag (~3.6% to ~7.2%) that erodes gross returns after fees. This finding aligns with transaction cost literature (Job, 2022; Alotaibi et al., 2022; Kevin & Yugopuspito, 2025), indicating that structural tail-risk architecture (CVaR) plays a central role in frictional emerging markets. It also qualifies the assertions of high-frequency quantitative models that advocate unconstrained algorithmic rebalancing, demonstrating that without explicit turnover penalties or execution smoothing bands, transaction friction rapidly neutralizes statistical predictive edges on frontier exchanges.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

1. Statistical Non-Normality and Tail-Risk Exposure: Formal statistical diagnostic tests (Jarque-Bera and Shapiro-Wilk) rejected Gaussian normality across all 28 evaluated NGX equities ($p < 0.001$). The empirical return distributions exhibited predominantly positive skewness (cross-sectional mean = +0.54) combined with excess kurtosis across all 28 assets (ranging from 2.20 to 14.47 among assets unaffected by corporate actions), indicating heavy-tailed, leptokurtic distributions that support the use of downside Conditional Value-at-Risk (CVaR) over standard variance.

2. Machine Learning Return Forecasting Dynamics: Non-linear machine learning ensembles (Random Forest and XGBoost) evaluated under Expanding-Window Walk-Forward Validation captured time-varying market dynamics. Random Forest achieved the highest average out-of-sample directional accuracy of 38.1% across liquid NGX equities (with individual assets reaching up to 50.8%), driven primarily by volume-confirmed momentum features (Percentage Price Oscillator, Relative Strength Index, and On-Balance Volume).

3. Out-of-Sample Portfolio Optimization Performance: Integrating machine learning return forecasts into a convex Mean-CVaR optimization framework generated higher risk-adjusted returns under zero fee friction relative to the market index. XGBoost + Mean-CVaR achieved a gross Sharpe ratio of 1.52 (42.00% annualized return, -22.09% maximum drawdown) compared to 0.91 for the passive NGX Index Buy-Hold baseline. This gross outperformance over the index is statistically significant at the unadjusted level (Ledoit-Wolf circular block bootstrap $p = 0.015$), but marginal after Benjamini–Hochberg False Discovery Rate correction for 18 simultaneous

comparisons ($q = 0.056$), while delivering Certainty Equivalent Return utility gains (+819 bps CER gain). Crucially, XGB-CVaR and Historical-CVaR achieved statistically indistinguishable gross Sharpe ratios (1.52 vs 1.50, difference +0.02, FDR $q = 0.591$), confirming that outperformance over the market index stems from the convex CVaR risk architecture rather than machine learning return forecasts. Under transaction friction, low-turnover Historical-CVaR significantly outperforms active XGB-CVaR under institutional fees ($q = 0.000$), and unconstrained MVO collapses against 1/N equal weighting under retail friction ($q = 0.009$).

4. Transaction Cost Dynamics and Turnover Friction: Evaluating portfolio performance under market stress and multi-tier transaction cost friction demonstrated that unconstrained Markowitz Mean-Variance Optimization experiences severe performance degradation (-86.38% drawdown, -64.09% CER utility) due to high turnover (99.28% weekly turnover). Under institutional (0.75%) and retail (1.50%) brokerage fees, active ML rebalancing turnover (11.00%) creates an annual fee drag (~3.6% to ~7.2%) that erodes gross returns after fees, while low-turnover Historical-CVaR (0.38% turnover) maintains stable post-fee performance (net Sharpe 1.48 to 1.49, pairwise Ledoit-Wolf $p = 0.000$, FDR $q = 0.000$ vs XGB-CVaR).

5.2 Conclusion

This study investigated equity return forecasting and tail-risk-aware portfolio optimization across 28 liquid equities on the Nigerian Exchange Group (NGX) spanning a 16-year period (2010-2025). The research evaluated non-normality diagnostics, Machine Learning return forecasting (Random Forest and XGBoost) under Expanding-Window Walk-Forward Validation, and convex Mean-CVaR asset allocation backtesting.

The overall conclusion of this research is that while tail-risk-aware CVaR optimization delivers substantial risk-adjusted outperformance and downside protection on the NGX (+819 bps

CER gain over passive indexing), machine learning return forecasts provide no statistically significant enhancement over simple historical sample means, and high-turnover rebalancing incurs severe transaction fee drag. Consequently, structural tail-risk management via Conditional Value-at-Risk (CVaR) appears to be the primary contributor to real-world investor economic surplus in the NGX context studied. This indicates that downside tail-risk architecture plays a far more critical role than model forecasting complexity in frictional emerging equity markets.

5.3 Policy and Practical Recommendations

5.3.1 Regulatory and Macroeconomic Policy Recommendations

1. **PENCOM Investment Guidelines Update:** The National Pension Commission (PENCOM) may consider incorporating downside tail-risk metrics, such as Conditional Value-at-Risk (CVaR at 95% confidence level), alongside traditional variance-based measures in its Investment Guidelines for Fund I, Fund II, and Fund III equity portfolios. Pension Fund Administrators (PFAs) could evaluate CVaR-constrained allocation models to protect pension assets during extreme macroeconomic shocks.

2. **SEC & NGX Regulatory Data Transparency:** The Securities and Exchange Commission (SEC) and NGX Regulation should establish open-access, low-latency API data infrastructure for market participants to support quantitative risk management. Furthermore, the SEC should require asset management firms to publish quarterly CVaR metrics in fund factsheets to enhance retail investor risk transparency.

5.3.2 Institutional Asset Allocation Recommendations

1. **Adoption of Low-Turnover Tail-Risk Frameworks:** Pension Fund Administrators (PFAs) and institutional fund managers operating on the NGX should evaluate Mean-CVaR asset allocation as a complement or alternative to classical Markowitz Mean-Variance Optimization. To

prevent fee erosion, institutional managers should enforce strict turnover caps, employ turnover-penalized objective functions, or adopt quarterly rebalancing protocols.

2. **Dynamic Risk-Free Asset Allocation:** Institutional portfolios should actively incorporate sovereign risk-free assets (such as Central Bank of Nigeria 91-day Treasury Bills) to stabilize portfolio Sharpe ratios during market drawdown regimes.

5.3.3 Quantitative Risk Management Recommendations

1. **Stress-Testing and Downside Risk Auditing:** Risk officers and quantitative portfolio managers should consider regular stress testing using historical block bootstrap resampling and non-parametric CVaR estimation to evaluate portfolio tail loss limits.

2. **Turnover-Constrained Rebalancing and Fee Auditing:** Quantitative portfolio managers on frontier exchanges must implement turnover constraints (such as rebalancing bands, penalty terms, or quarterly re-allocation schedules) to prevent aggressive portfolio turnover from eroding gross risk-adjusted performance across institutional and retail transaction fee tiers.

5.4 Limitations of the Study

While this study provides empirical insights into quantitative asset management on the Nigerian equity market, several methodological limitations are acknowledged:

1. **Asset Universe Scope:** The study evaluated 28 liquid equities from the NGX Pension Index with complete 16-year histories (2010–2025). Recently listed high-capitalization assets (e.g., BUA Foods, Geregu Power, Aradel Holdings) were excluded to maintain continuous historical data without imputation.

2. **Data Frequency:** Analysis was conducted using weekly log returns. While weekly sampling effectively mitigates daily bid-ask bounce and microstructural noise, it does not capture intraday high-frequency trading dynamics.

3. **Exogenous Macroeconomic Signals:** The machine learning feature matrix focused on price and volume technical indicators; exogenous macroeconomic variables (such as USD/NGN exchange rate shocks and Brent crude oil prices) were not explicitly included in the feature set.

4. **Sample Selection Bias:** The requirement for complete 16-year price histories (2010–2025) may introduce survivorship and sample-selection bias, as only securities that remained continuously listed and traded throughout the full sample period were retained. Firms that delisted, were suspended, or entered the market after 2010 are not represented in the analysis. This limitation should be considered when generalizing findings to the broader NGX universe.

5. **Price-Only Returns and Unadjusted Corporate Actions:** This study used price-only (capital appreciation) returns rather than total returns inclusive of dividend distributions. Given that several NGX equities offer meaningful dividend yields, the reported returns may understate total investment performance across all strategies. Additionally, prices were not adjusted for corporate actions; these observations reflect mechanical price changes rather than economic returns and inflate the reported kurtosis for these three assets (MANSARD, TRANSCORP, and WEMABANK; see Section 4.2).

5.5 Suggestions for Further Research

Based on the empirical findings and limitations of this research, the following directions for future investigation are suggested:

1. **Macroeconomic and Exogenous Feature Integration:** Future research should expand the machine learning feature space to include macroeconomic variables, foreign exchange volatility, and crude oil price dynamics.

2. Deep Learning and High-Frequency Architectures: Extending forecasting models to Transformer-based temporal models and Long Short-Term Memory (LSTM) networks on daily or intraday NGX trading data.

3. Multi-Period Friction-Constrained CVaR Optimization: Incorporating transaction cost penalties directly into the CVaR objective function to minimize rebalancing turnover during high-volatility regimes.

5.6 Contribution to Knowledge

This research contributes to quantitative finance literature in the following ways:

1. Empirical Contribution: Provides an empirical evaluation of Machine Learning integrated with convex Mean-CVaR optimization on the Nigerian Exchange Group using a clean 16-year dataset (2010–2025).

2. Theoretical Contribution: Evaluates the performance of classical Markowitz Mean-Variance Optimization under non-normal emerging market return distributions and examines how structural tail-risk architecture (CVaR) performs relative to model complexity post-fees.

3. Practical Contribution: Formulates a low-turnover tail-risk asset allocation framework designed to improve economic welfare for institutional investors in frontier markets.

5.7 Data and Code Availability Statement

To guarantee complete computational reproducibility and scientific transparency, the entire quantitative finance pipeline, encompassing data cleaning scripts, non-normality diagnostic tests, technical feature extraction (with Welles Wilder's Parabolic SAR), expanding walk-forward machine learning models (Random Forest and XGBoost with TimeSeriesSplit cross-validation tuning), convex Mean-CVaR linear programming portfolio optimization backtests, is open-source and publicly hosted on GitHub at:

Repository URL: <https://github.com/abdulameen962/equity-optimization>

REFERENCES

- Adegboyo, O. S., & Sarwar, K. (2025). Modelling and forecasting of Nigeria stock market volatility. *Future Business Journal*, 11(1), Article 124. <https://doi.org/10.1186/s43093-025-00536-4>
- Ajiga, D. I., Adeleye, R. A., Tubokirifuruar, T. S., Bello, B. G., Ndubuisi, N. L., Asuzu, O. F., & Owolabi, O. R. (2024). Machine learning for stock market forecasting: A review of models and accuracy. *Finance & Accounting Research Journal*, 6(2).
- Alfeus, M., Harvey, J., & Maphatsoe, P. (2024). Improving realised volatility forecast for emerging markets. *Journal of Economics and Finance*. Advance online publication. <https://doi.org/10.1007/s12197-024-09701-x>
- Alfzari, S., Al-Shboul, M., & Alshurideh, M. (2025). Predictive analytics in portfolio management: A fusion of AI and investment economics for optimal risk-return trade-offs. *International Review of Management and Marketing*, 15(2), 365–380. <https://doi.org/10.32479/irmm.18594>
- Alim, W., Khan, N. U., Zhang, V. W., Cai, H. H., Mikhaylov, A., & Yuan, Q. (2024). Influence of political stability on the stock market returns and volatility: GARCH and EGARCH approach. *Financial Innovation*. <https://doi.org/10.1186/s40854-024-00658-8>
- Alotaibi, T. S., Dalla Valle, L., & Craven, M. J. (2022). The worst case GARCH-copula CVaR approach for portfolio optimisation: Evidence from financial markets. *Journal of Risk and Financial Management*, 15(10), Article 482. <https://doi.org/10.3390/jrfm15100482>
- Arif, U., Sohail, M. T., & Majeed, M. I. (2020). Portfolio optimization with mean-variance & mean-CVaR: Evidence from Pakistan stock market. *International Journal of Management Research & Emerging Sciences*, 10(2), 215–226.

- Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228. <https://doi.org/10.1111/1467-9965.00068>
- Ashrafzadeh, M., Sadrani, M., & Zolfani, S. H. (2025). Clustering-based return prediction model for stock pre-selection in portfolio optimization. *Results in Engineering*, 27, Article 106263. <https://doi.org/10.1016/j.rineng.2025.106263>
- Bali, T. G., Gokcan, S., & Liang, B. (2007). Value at risk and the cross-section of hedge fund returns. *Journal of Banking & Finance*, 31(4), 1135–1166. <https://doi.org/10.1016/j.jbankfin.2006.10.023>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Bodnar, T., Lindholm, M., Niklasson, V., & Thorsén, E. (2022). Bayesian portfolio selection using VaR and CVaR. *Applied Mathematics and Computation*, 427, Article 127120.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Chao, L. (2024). Application of machine learning in stock market return forecasting. *International Journal of Scientific Research and Management (IJSRM)*, 12(7), 6827–6835. <https://doi.org/10.18535/ijssrm/v12i07.em11>
- Chaweewanchon, A., & Chaysiri, R. (2022). Markowitz mean-variance portfolio optimization with predictive stock selection using machine learning. *International Journal of Financial Studies*, 10(3), Article 64. <https://doi.org/10.3390/ijfs10030064>

- Chen, R. (2025). Stock price prediction and portfolio optimization based on mean variance model and random forest model. *Advances in Economics, Business and Management Research*, 333, 358–367.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Cheng, Y. (2025). Monte Carlo-Based VaR Estimation and Backtesting Under Basel III. *Risks*, 13(8), Article 146. <https://doi.org/10.3390/risks13080146>
- Chung, V., Espinoza, J., & Quispe, R. (2025). Forecasting Financial Volatility Under Structural Breaks: A Comparative Study of GARCH Models and Deep Learning Techniques. *Journal of Risk and Financial Management*, 18(9), Article 494. <https://doi.org/10.3390/jrfm18090494>
- DeMiguel, V., Garlappi, L., & Uppal, R. (2009). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5), 1915–1953. <https://doi.org/10.1093/rfs/hhm075>
- Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431. <https://doi.org/10.2307/2286348>
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. <https://doi.org/10.1080/07350015.1995.10524599>
- Espiga-Fernández, F., García-Sánchez, Á., & Ordieres-Meré, J. (2024). A Systematic Approach to Portfolio Optimization: A Comparative Study of Reinforcement Learning Agents,

- Market Signals, and Investment Horizons. *Algorithms*, 17(12), Article 570.
<https://doi.org/10.3390/a17120570>
- Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2), 383–417. <https://doi.org/10.1111/j.1540-6261.1970.tb00518.x>
- Fan, Y. (2025). Enhancing investment strategies with LSTM-based stock prediction and mean-variance portfolio optimization. *Advances in Economics, Management and Political Sciences*, 182, 110–117. <https://doi.org/10.54254/2754-1169/2024.23667>
- Fapetu, O., Ojo, S. M., Balogun, A. A., & Asaolu, A. A. (2021). Capital market performance and macroeconomic dynamics in Nigeria. *FUOYE Journal of Finance and Contemporary Issues*, 1(1), 29–37.
- Fatouros, G., Makridis, G., Kotios, D., Soldatos, J., Filippakis, M., & Kyriazis, D. (2023). DeepVaR: A framework for portfolio risk assessment leveraging probabilistic deep neural networks. *Digital Finance*, 5(1), 29–56.
- Ferrari, D., Paterlini, S., Rigamonti, A., & Weissensteiner, A. (2024). Smoothed semicovariance estimation for portfolio selection. *Annals of Operations Research*. Advance online publication. <https://doi.org/10.1007/s10479-024-06043-z>
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5), 2223–2273. <https://doi.org/10.1093/rfs/hhz113>
- He, W. (2022). An empirical research based on Markowitz's portfolio theory. *World Scientific Research Journal*, 8(3), 221–227. [https://doi.org/10.6911/WSRJ.202203_8\(3\).0028](https://doi.org/10.6911/WSRJ.202203_8(3).0028)
- Hsiao, Y.-Y. (2025). A comprehensive survey of modern portfolio optimization: From traditional risk analysis to advanced analytics and machine learning approaches. *Advances in*

- Economics, Management and Political Sciences, 166, 189–194.
<https://doi.org/10.54254/2754-1169/2025.21140>
- Huang, R., Kambouroudis, D., & McMillan, D. G. (2025). Is portfolio diversification still effective: Evidence spanning three crises from the perspective of U.S. investors. *Journal of Asset Management*, 26(2), 115–135. <https://doi.org/10.1057/s41260-025-00398-z>
- Jarque, C. M., & Bera, A. K. (1987). A test for normality of observations and regression residuals. *International Statistical Review*, 55(2), 163–172.
<https://doi.org/10.2307/1403192>
- Job, O. D. (2022). An empirical evaluation of alternative asset allocation policies for emerging and frontier market investors in Africa. *Journal of Financial Risk Management*, 11(3), 481–521. <https://doi.org/10.4236/jfrm.2022.113024>
- Jobson, J. D., & Korkie, B. M. (1981). Performance hypothesis testing with the Sharpe and Treynor measures. *The Journal of Finance*, 36(4), 889–908.
- Kevin, J., & Yugopuspito, P. (2025). Hybrid LSTM and PPO networks for dynamic portfolio optimization. arXiv:2511.17963. <https://arxiv.org/abs/2511.17963>
- Leccadito, A., Staino, A., & Toscano, P. (2024). A novel robust method for estimating the covariance matrix of financial returns with applications to risk management. *Financial Innovation*, 10, Article 116.
- Ledoit, O., & Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Li, G. (2023). Portfolio optimization and risk analysis in financial markets. *Proceedings of the 2nd International Conference on Financial Technology and Business Analysis*, 61, 236–246. <https://doi.org/10.54254/2754-1169/61/20231273>

- Lorimer, D. A., van Schalkwyk, C. H., & Szczygieski, J. J. (2024). Portfolio optimisation using alternative risk measures. *Finance Research Letters*, 67, Article 105758.
<https://doi.org/10.1016/j.frl.2024.105758>
- Lyu, X. (2024). Portfolio Optimization Strategies: New Approaches Based on Machine Learning Forecasting. *Highlights in Business, Economics and Management*, 40, 1077–1082.
- Mahadevaswamy, G. H., & Shyamala, G. (2022). Markowitz Model Is Right Choice to Investment. *International Journal of Novel Research and Development*, 7(7), 305–314.
- Manogna, R. L., & Kulkarni, N. (2025). Portfolio Optimization Model for Stock Price Prediction Using Machine Learning. *Journal of Statistical Theory and Applications*, 24, 1091–1108.
<https://doi.org/10.1007/s44199-025-00140-z>
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91.
<https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>
- Martínez-Barbero, X., Cervelló-Royo, R., & Ribal, J. (2024). Portfolio optimization with prediction-based return using Long Short-Term Memory neural networks: Testing on upward and downward European markets. *Computational Economics*, 65, 1479–1504.
<https://doi.org/10.1007/s10614-024-10604-6>
- Mba, J. C., Ababio, K. A., & Agyei, S. K. (2022). Markowitz mean-variance portfolio selection and optimization under a behavioral spectacle: New empirical evidence. *International Journal of Financial Studies*, 10(2), Article 28. <https://doi.org/10.3390/ijfs10020028>
- Memmel, C. (2003). Performance hypothesis testing with the Sharpe ratio. *Finance Letters*, 1(1), 21–23.
- Michaud, R. O. (1989). The Markowitz optimization enigma: Is 'optimized' optimal? *Financial Analysts Journal*, 45(1), 31–42. <https://doi.org/10.2469/faj.v45.n1.31>

- Moyoweshumba, E., & Seitshiro, M. (2025). Leveraging Markowitz, Random Forest, and XGBoost for optimal diversification of South African stock portfolios. *Data Science in Finance and Economics*, 5(2), 205–233. <https://doi.org/10.3934/DSFE.2025010>
- Mozumder, S., Hasan, M. K., & Kabir, M. H. (2024). An evaluation of the adequacy of Lévy and extreme value tail risk estimates. *Financial Innovation*, 10(1), Article 100. <https://doi.org/10.1186/s40854-024-00614-6>
- Naeem, M., Jassim, H. S., & Korsah, D. (2024). The application of machine learning techniques to predict stock market crises in Africa. *Journal of Risk and Financial Management*, 17(12), Article 554. <https://doi.org/10.3390/jrfm17120554>
- Nahari, F. (2025). Portfolio optimization in practice: A comparative analysis of the Markowitz and Index models. In M. M. Husin (Ed.), *Proceedings of the 2025 International Conference on Financial Risk and Investment Management (ICFRIM 2025)*, *Advances in Economics, Business and Management Research* (Vol. 333, pp. 367–375). Atlantis Press.
- Nigerian Exchange Group. (2024). Market report and listed securities directory. NGX Group. <https://ngxgroup.com>
- Ojo, A. K., & Okafor, I. J. (2024). Forecasting Nigerian Equity Stock Returns Using Long Short-Term Memory Technique. *Journal of Advances in Mathematics and Computer Science*, 39(7), 45–54. <https://doi.org/10.9734/jamcs/2024/v39i71911>
- Okafor, C., & Robertson, A. (2023). Maximizing returns: Portfolio optimization in the Nigerian Stock Exchange. *International Journal of Advances in Applied Mathematics and Computer Science*, 10(2), 14–28. <https://americaserial.com/journals/ijaamcs/article/518>

- Rigamonti, A., & Lučivjanská, K. (2024). Mean-semivariance portfolio optimization using minimum average partial. *Annals of Operations Research*, 334, 185–203.
<https://doi.org/10.1007/s10479-022-04736-x>
- Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41. <https://doi.org/10.21314/JOR.2000.038>
- Rockafellar, R. T., & Uryasev, S. (2002). Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7), 1443–1471. [https://doi.org/10.1016/S0378-4266\(02\)00271-6](https://doi.org/10.1016/S0378-4266(02)00271-6)
- Sahiner, M. (2022). Forecasting volatility in Asian financial markets: Evidence from recursive and rolling window methods. *SN Business & Economics*, 2, Article 157.
<https://doi.org/10.1007/s43546-022-00329-9>
- Salo, A., Doumpos, M., Liesiö, J., & Zopounidis, C. (2024). Fifty years of portfolio optimization. *European Journal of Operational Research*, 318(1), 1–18.
<https://doi.org/10.1016/j.ejor.2023.12.031>
- Samaniego Alcántar, A. (2023). Semi-variance optimization for the components of the Dow Jones Industrial Average index. *Contaduría y Administración*, 68(4), 1–17.
<http://dx.doi.org/10.22201/fca.24488410e.2023.3409>
- San, J. (2025). Stock Forecasting and Portfolio Optimization Based on ARIMA-GARCH, Random Forest and Monte Carlo Models. *Advances in Economics, Business and Management Research*, 333, 582–591. https://doi.org/10.2991/978-94-6463-652-9_61
- Senescall, M., & Low, R. K. Y. (2024). Quantitative Portfolio Management: Review and Outlook. *Mathematics*, 12(18), Article 2897. <https://doi.org/10.3390/math12182897>

- Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3/4), 591–611. <https://doi.org/10.2307/2333709>
- Sharpe, W. F. (1966). Mutual fund performance. *The Journal of Business*, 39(1), 119–138. <https://doi.org/10.1086/294846>
- Ślusarczyk, D., & Ślepaczuk, R. (2025). Optimal Markowitz portfolio using returns forecasted with time series and machine learning models. *Journal of Big Data*, 12, Article 127. <https://doi.org/10.1186/s40537-025-01164-z>
- Sortino, F. A., & Price, L. N. (1994). Performance measurement in a downside risk framework. *The Journal of Investing*, 3(3), 59–64. <https://doi.org/10.3905/joi.3.3.59>
- Sulaiman, L. A., Adejayan, A. O., & Ilori, O. O. (2023). Capital Market Development and Economic Growth of West African Countries. *Nigerian Journal of Banking and Financial Issues*, 9(1), 117–125.
- Tjiwidjaja, H. (2025). Optimization of Investment Portfolio Returns Through an Integrated Risk Management Approach. *STIE Ganesha Research Papers*, 853–863.
- Uzoaga, G. A., Adenomon, M. O., Nweze, N. O., & Maijama, B. (2025a). Modelling and predicting stock prices of Nigerian Stock Exchange using some machine learning techniques and time series model. *Science World Journal*, 20(2), 510–515. <https://dx.doi.org/10.4314/swj.v20i2.9>
- Uzoaga, G. A., Adenomon, M. O., Nweze, N. O., & Maijamaa, B. (2025b). Predictive machine learning methods for stock returns among emerging economies in Africa. *Science World Journal*, 20(3), 941–947. <https://dx.doi.org/10.4314/swj.v20i3.3>
- Wahid, A. J., Riaman, & Sukono. (2025). A Systematic Literature Review on Mean-CVaR Based Financial Asset Portfolio Weight Allocation Using K-Means Clustering.

- CAUCHY – Jurnal Matematika Murni dan Aplikasi, 10(2), 1069–1091.
<https://doi.org/10.18860/cauchy.v10i2.36590>
- Wang, T., Pan, Q., Wu, W., Gao, J., & Zhou, K. (2024). Dynamic Mean–Variance Portfolio Optimization with Value-at-Risk Constraint in Continuous Time. *Mathematics*, 12(14), Article 2268. <https://doi.org/10.3390/math12142268>
- Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6), 80–83. <https://doi.org/10.2307/3001968>
- Yadav, A., Madhavi, R., Bagaria, O., Ambulkar, A., & Sharma, S. (2024). Survey on financial portfolio management's role in investment decision-making strategies. *Multidisciplinary Reviews*, 6, Article e2023ss101. <https://doi.org/10.31893/multirev.2023ss101>
- Yu, S. (2024). Advancing stock market return forecasting with LSTM models and financial indicators. In *Proceedings of the International Conference on Economic Management and Green Development, Advances in Economics, Management and Political Sciences* (Vol. 122, pp. 137–144). <https://doi.org/10.54254/2754-1169/2024.17734>
- Zaimovic, A., Arnaut-Berilo, A., & Bešlija, R. (2024). International portfolio diversification benefits: An empirical investigation of the 28 European stock markets. *The South East European Journal of Economics and Business*, 19(1), 96–112. <https://doi.org/10.2478/jeb-2024-0007>
- Zapata Quimbayo, C., & León, B. (2025). Downside risk measures and ESG factors in optimal portfolio construction: Evidence from European equity markets. *Economics - Innovative and Economics Research Journal*, 13(4), 5–17. <https://doi.org/10.2478/eoik-2025-0082>

- Zhang, G. (2025). Using machine learning for stock return prediction. *Advances in Economics, Management and Political Sciences*, 185, 119–126. <https://doi.org/10.54254/2754-1169/2025.LH23915>
- Zhang, Y. (2024). Integrating forecasting and mean-variance portfolio optimization: A machine learning approach. *Advances in Economics, Management and Political Sciences*, 90, 157–168. <https://doi.org/10.54254/2754-1169/90/20242002>
- Zsurkis, G., Nicolau, J., & Rodrigues, P. M. M. (2024). First passage times in portfolio optimization: A novel nonparametric approach. *European Journal of Operational Research*, 312(3), 1074–1085. <https://doi.org/10.1016/j.ejor.2023.07.044>